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An advanced framework for mobile robot trajectory tracking controller via recurrent general Type-2 fuzzy neural networks

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ABSTRACT

This paper presents an advanced framework for mobile robot trajectory tracking based on a recurrent general type-2 Takagi-Sugeno-Kang fuzzy neural network (RGT2TSKFNN). The proposed framework addresses limitations of existing methods by enhancing adaptability, robustness, and precision under uncertain conditions. The trajectory tracking controller comprises two components: an adaptive kinematic controller that employs a cost function and specific constraints to guide the mobile robot along a desired reference trajectory, and a dynamic controller constructed using the novel RGT2TSKFNN architecture. Both the network structure and parameters are updated online, with parameter adaptation governed by a Lyapunov-based learning law to guarantee stability. The efficacy of the proposed approach is demonstrated through comprehensive simulations and real-world experiments under uncertainty, highlighting its ability to achieve accurate and reliable trajectory tracking. Comparative results indicate that the proposed controller outperforms existing methods, yielding lower performance indices such as mean absolute error, integral of absolute error, and integral of squared error across both simulations and experiments.

1. Introduction

Mobile robots have emerged as a fundamental component of modern automation, with applications encompassing industrial automation, healthcare, agriculture, and autonomous vehicles. Their capability to navigate dynamic environments, avoid obstacles, and execute precise tasks has rendered them indispensable in both structured and unstructured contexts (Zhang et al., 2018; Rubio et al., 2019; Alshorman et al., 2020; Ren et al., 2020; Niloy et al., 2021). A pivotal aspect of mobile robot performance is trajectory tracking, which refers to the ability of the robot to follow a predefined path with high accuracy and robustness. Achieving precise trajectory tracking presents significant challenges due to uncertainties in the robot's dynamics, external disturbances, and environmental variability (Xiao et al., 2024). Traditional control methodologies, such as proportional-integral-derivative (PID) controller frequently encounter difficulties in managing these complexities, thereby necessitating the development of advanced control frameworks (Liu et al., 2008; Rossomando et al., El-Nagar and El-Bardini, 2014; 2015; Hanna et al., 2023; Huang and Gao, 2024).

In recent years, intelligent control systems have gained prominence for their efficiency in modeling complex, nonlinear systems and adapting to uncertainties (Abdelsattar et al., 2024, 2025). Among these, fuzzy logic systems (FLSs) and neural networks (NNs) have demonstrated considerable potential in addressing the challenges associated with trajectory tracking (Rossomando et al., 2015; Hassan and Saleem, 2022; El-Nagar et al., 2025; Hanna et al., 2025). FLSs excel in processing imprecise and uncertain information, while NNs provide robust learning capabilities for modeling nonlinear dynamics. The integration of these methodologies, particularly through the implementation of fuzzy neural networks (FNNs), has led to significant advancements in mobile robot control (Khater et al., 2025). However, conventional FNNs are often constrained by their reliance on type-1 fuzzy sets, which are insufficient

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to fully encapsulate the high levels of uncertainty inherent in real-world environments, so type-2 FNN is presented in (Zhao et al., 2024) for solving the uncertainty problems.

1.1. Literature review

The application of FLS in mobile robots has been the focus of extensive research over the past few decades. Fuzzy logic, introduced by Zadeh, offers a mathematical framework for managing imprecise and uncertain information, rendering it particularly effective for controlling complex, nonlinear systems such as mobile robots (El Sayed et al., 2024). This section reviews the evolution of fuzzy logic-based approaches in mobile robot control, emphasizing trajectory tracking, navigation, and obstacle avoidance.

Castillo et al. (2015) developed a type-1 fuzzy logic controller (T1-FLC) for a mobile robot trajectory tracking, incorporating membership function optimization through the ant colony algorithm. Similarly, Li et al. (2003) proposed another T1-FLC that utilized a rule-based mapping from sensor inputs to control outputs, demonstrating satisfactory performance in structured environments. However, T1-FLCs exhibit limitations in managing high levels of uncertainty, prompting the exploration of type-2 fuzzy logic systems (T2-FLS) as a more robust alternative.

Consequently, interval type-2 fuzzy logic controllers (IT2-FLCs) represent a viable alternative to address the limitations of T1-FLCs (Al-Mahturi et al., 2020). IT2-FLCs have been suggested for mobile robot control applications, wherein they have demonstrated superior tracking performance compared to their type-1 counterparts (Huang et al., 2017; Lu et al., 2018; Al-Mahturi et al., 2021). However, conventional IT2-FLCs typically feature static structures and lack the ability to adapt their configuration in real time. The generation of type-2 fuzzy rules and their corresponding membership functions presents a significant challenge, particularly for systems with numerous variables (Al-Mahturi et al., 2023). To address this, the development of interval type-2 fuzzy neural networks (IT2FNNs) is essential, as they enable self-learning capabilities either from scratch or by refining a predefined rule base.

IT2FNN gets their learning ability from NNs, and their inference technology from type-2 fuzzy systems. As such, IT2FNN are an effective tool for mobile robot control. These networks typically include the adaptive network-based type-2 fuzzy inference system, which employs either a Takagi–Sugeno–Kang (TSK) type consequent represented by a linear equation or a Mamdani-type consequent (Khater et al., 2019a, 2019b, 2020). For example, Kim and Chwa (2014) developed an IT2FNN for obstacle avoidance in wheeled mobile robots. Their results showed that the IT2FNN-based controller outperformed its type-1 fuzzy counterpart by demonstrating increased robustness to uncertainties during obstacle navigation. To further enhance performance, Bencherif and Chouireb (2019) proposed a self-evolving recurrent TSK interval type-2 fuzzy neural network (RTSKIT2FNN) specifically for mobile robot trajectory tracking. The self-evolving algorithm enables the RTSKIT2FNN to adapt its network structure dynamically during online learning, while a varied learning rates algorithm is employed to optimize the gradient descent process. This combination ensures stable convergence of the control system, as rigorously demonstrated through a discrete Lyapunov stability analysis. The proposed approach exhibits improved performance relative to several previously published methods, including various type-1 and (IT2FNNs) such as the adaptive network based fuzzy inference system (ANFIS) (Aissa and Fatima, 2015), recurrent fuzzy neural network (RFNN) (Mon et al., 2008), IT2FNN with fixed structure using sliding mode (IT2FNN-SM) (Kayacan and Maslim, 2016), IT2FNN (Abiyev and Kaynak, 2010), self-organization T2FNN using gradient descent for parameters learning and the particle swarm optimization (Lin and Le, 2017), self-evolving IT2FNN (SEIT2FNN) (Lin et al., 2018) and simplified IT2FNN (SIT2FNN) (Lin et al., 2013). Nevertheless, the algorithm presented by Bencherif and Chouireb (2019) was validated only on a single trajectory tracking task and has yet to be evaluated under conditions involving uncertainties.

Despite significant advancements in fuzzy neural network-based controllers for mobile robots, several limitations persist in the existing literature. Many approaches, including IT2FNN and RTSKIT2FNN models, often rely on fixed or partially adaptive structures and have been validated primarily under ideal or limited scenarios, such as single trajectory tracking tasks. Moreover, these studies frequently overlook the impact of real-world uncertainties, such as variable disturbances, parameter variations, and measurement errors. Consequently, there is a clear need for adaptive, computationally efficient controllers that are rigorously tested under diverse uncertainty conditions to bridge these gaps and enhance the practical applicability of fuzzy neural network controllers in mobile robotics.

1.2. Motivation

The motivation for this research stems from the growing need for mobile robots capable of operating reliably in complex and uncertain environments. Applications such as autonomous vehicles, warehouse automation, and search-and-rescue missions demand high levels of adaptability and precision in trajectory tracking under dynamic and unpredictable conditions. While traditional control methods have demonstrated effectiveness in structured settings, they often prove inadequate in such scenarios due to their limited ability to manage system uncertainties, external disturbances, and nonlinear dynamics.

These limitations highlight the need for advanced control frameworks capable of delivering robust and adaptive performance in uncertain and nonlinear environments. General type-2 fuzzy sets (GT2FSs) present a promising solution by offering greater flexibility in modeling uncertainties. When integrated with TSK fuzzy systems and recurrent neural networks (RNNs), GT2FSs facilitates the development of control systems that are both interpretable and capable of learning complex dynamics. This integration serves as the foundation for the proposed recurrent general type-2 TSK fuzzy neural network (RGT2TSKFNN), specifically designed to tackle the challenges associated with mobile robot trajectory tracking. In the proposed framework, structure learning is utilized to automatically determine the appropriate number of fuzzy rules, while parameter learning is guided by update laws derived from Lyapunov theory, ensuring convergence and system stability. In this study, the adaptive dynamic controller is designed based on the proposed RGT2TSKFNN. Additionally, the parameters for the kinematic controller are adjusted online to enhance the robustness and improve the overall performance.

1.3. Novelties and contributions

The primary contributions of this paper are summarized as follows:

1. *Enhanced robustness*: The proposed framework leverages GT2FSs to significantly improve its capacity to mitigate the effect of uncertainties, thereby ensuring reliable performance in dynamic environments, an essential requirement for mobile robotics. Additionally, the integration of recurrent TSK FNNs enhances the performance, enabling for real-time adjustments to change conditions, which are vital for operations in unpredictable scenarios.
2. *Enhanced performance through adaptive kinematic control*: The framework includes an adaptive kinematic control mechanism that improves overall performance, leading to superior trajectory tracking capabilities compared to traditional methods.
3. *Novel dynamic controller structure with reduction in rule complexity*: A dynamic controller structure based on parameter learning, derived from the Lyapunov function, is introduced to ensure system stability. This innovative approach effectively addresses a fundamental challenge in control system design. Furthermore, reducing the number of rules required compared to recurrent IT2FNNs, the proposed

framework simplifies control logic and enhances computational efficiency.

4. *Advancement of mobile robots*: By addressing the limitations of existing trajectory tracking methods, the proposed framework significantly advances the state of the art in mobile robotic applications, contributing to the development of more capable and reliable robotic systems for real-world applications.

The remainder of this paper is organized as follows: Section 2 presents the modeling of the mobile robot. Section 3 details the proposed RGT2TSKFNN structure. Section 4 discusses the online learning for the proposed RGT2TSKFNN. Section 5 presents the design of the trajectory tracking controller, which includes both the adaptive kinematic controller and the adaptive dynamic controller. Section 6 illustrates the application of the algorithm through the simulation results. Section 7 presents the experimental results followed by the discussion, conclusions, and references.

2. Modeling of mobile robot

Fig. 1 shows the structure of a non-holonomic mobile robot and its parameters and variables of interest in the inertial cartesian plane $X-Y$, where x and y are the position in the $X-Y$ plane. It consists of two drive wheels mounted on the same axis and a free castor wheel. The two wheels receive the control commands, which include the linear velocity v_r and angular velocity ω_r. The robot orientation is denoted as θ_r, while the mass center of the robot is denoted as G. C denotes the castor wheel position, c is the distance between the axis of the traction wheels and the free castor wheel and a_r is the distance between the axis of the reaction wheels and the G point. The mathematical model of the mobile robot is divided into two models, which are the kinematic and dynamic models. It can be given as follows (Rossomando et al., 2015):

2.1. Kinematic model

The mathematical equation for the kinematic model in discrete form is given as (Rossomando et al., 2015):

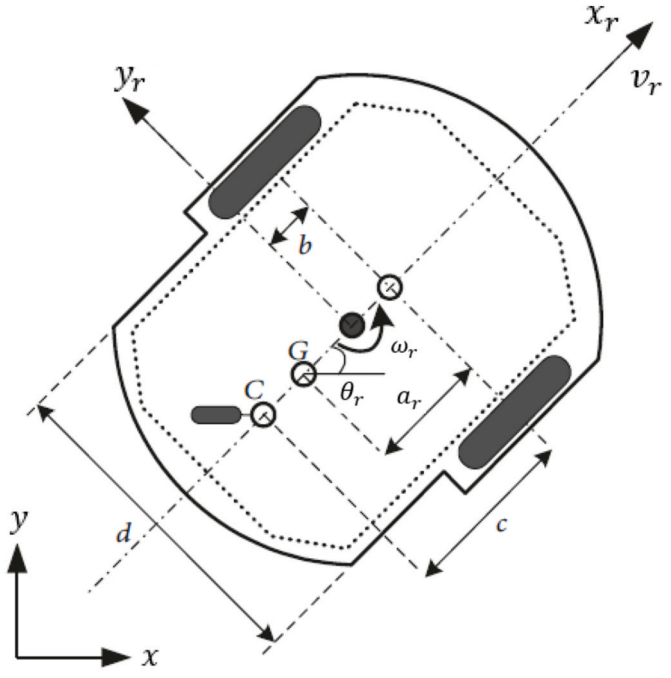


Fig. 1. Mobile robot parameters and variables.

$$\begin{bmatrix} x(\tau) \\ y(\tau) \\ \theta_r(\tau) \end{bmatrix} = T \begin{bmatrix} \cos\theta_r(\tau) & -a_r\sin\theta_r(\tau) \\ \sin\theta_r(\tau) & a_r\cos\theta_r(\tau) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_r(\tau) \\ \omega_r(\tau) \end{bmatrix} + \begin{bmatrix} x(\tau-1) \\ y(\tau-1) \\ \theta_r(\tau-1) \end{bmatrix} + \begin{bmatrix} \rho_x \\ \rho_y \\ 0 \end{bmatrix} \quad (1)$$

where τ represents the iteration number, T is the sampling period and ρ_x and ρ_y are functions of slip velocities and orientation and, which considered disturbances.

2.2. Dynamic model

The mathematical equation for the dynamic model in discrete form is given as (Rossomando et al., 2015):

$$\begin{bmatrix} v_r(\tau) \\ \omega_r(\tau) \end{bmatrix} = \begin{bmatrix} \frac{\lambda_3}{\lambda_1}\omega_r(\tau-1)^2 + \frac{\lambda_4}{\lambda_1}v_r(\tau-1) \\ -\frac{\lambda_5}{\lambda_2}v_r(\tau-1)\,\omega_r(\tau-1) + \frac{\lambda_6}{\lambda_2}\omega_r(\tau-1) \end{bmatrix} + \begin{bmatrix} \frac{1}{\lambda_1} & 0 \\ 0 & \frac{1}{\lambda_2} \end{bmatrix} \begin{bmatrix} v_{ref}(\tau) \\ \omega_{ref}(\tau) \end{bmatrix} + \begin{bmatrix} \rho_v \\ \rho_\omega \end{bmatrix} \quad (2)$$

where $v_{ref}(\tau)$ and $\omega_{ref}(\tau)$ are the desired linear and angular velocities respectively, which represent the control action. ρ_v and ρ_ω are functions of the physical mobile robot parameters as mass, inertia, wheel and tires diameters, motor and its parameters, forces on the wheels, and others. In this article, both ρ_v and ρ_ω are considered as disturbances. The dynamic model parameters are defined as:

$$\frac{\lambda_3}{\lambda_1} = T\frac{\zeta_3}{\zeta_1};\ \frac{\lambda_4}{\lambda_1} = \left(1 - T\frac{\zeta_4}{\zeta_1}\right);\ \frac{1}{\lambda_1} = T\frac{1}{\zeta_1};\ \frac{\lambda_5}{\lambda_2} = -T\frac{\zeta_5}{\zeta_2};\ \frac{\lambda_6}{\lambda_2} = \left(1 - T\frac{\zeta_6}{\zeta_2}\right);\ \frac{1}{\lambda_2} = T\frac{1}{\zeta_2} \quad (3)$$

The identified parameters for the Pioneer DX3 mobile robot, utilized in this study are obtained as (Capraro et al., 2017); $\zeta_1 = 0.24089$, $\zeta_2 = 0.2424$, $\zeta_3 = -0.000936$, $\zeta_4 = 0.9962$, $\zeta_5 = -0.00572$ and $\zeta_6 = 1.0915$.

3. Proposed RGT2TSKFNN structure

In this section, the structure of the proposed RGT2TSKFNN is introduced, which features multiple inputs and a single output. The proposed architecture consists of seven layers, as illustrated in Fig. 2. It employs GT2FSs in the antecedent portion and Takagi-Sugeno-Kang (TSK) type fuzzy rules in the consequent portion. Each rule for the RGT2TSKFNN is defined as follows:

$R_\ell^{\alpha_\vartheta}$: If $\upsilon_1(\tau)$ is $\widetilde{H}_1^\ell$ and…$\upsilon_\chi(\tau)$ is $\widetilde{H}_\chi^\ell$ Then

$$u_\ell^{\alpha_\vartheta}(\tau+1) \text{ is } \widetilde{\mathscr{G}}_0^\ell + \sum_{k=1}^{\chi} \widetilde{\mathscr{G}}_k^\ell\, \upsilon_k(\tau), \ell = 1,..,P \quad (4)$$

where $\upsilon_1(\tau),\ldots,\upsilon_\chi(\tau)$ denote the external inputs, with χ indicating the total number of inputs. $\widetilde{H}_k^\ell$ represents GT2FSs. P signifies the number of rules, while $\widetilde{\mathscr{G}}_k^\ell$ is an interval set defined as $\widetilde{\mathscr{G}}_k^\ell = \left[a_k^\ell - b_k^\ell, a_k^\ell + b_k^\ell\right]$ where the parameters a_k^ℓ, and b_k^ℓ correspond to the center and spread of the kth input coefficient in rule ℓ, respectively. The output at each layer n is represented by $\psi_\ell^{(n)}$. The explanation of each layer in Fig. 2 are presented in detail as follows:

Layer 1 (Input layer): In this layer, only the immediate states of the identified plant are provided as inputs to the network.

Layer 2 (Membership layer): This layer calculates a recurrent general type-2 membership function gradients to carry out the fuzzification

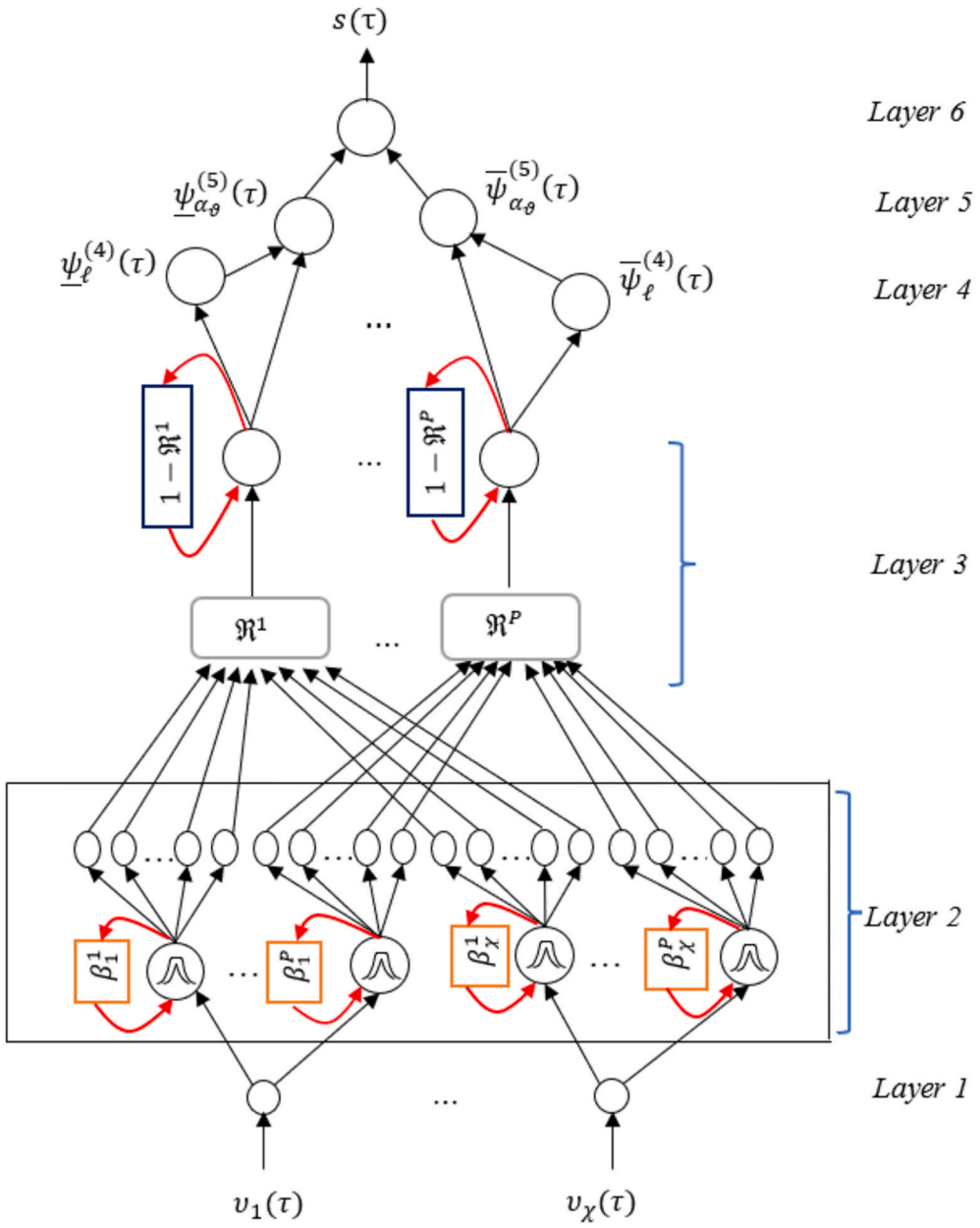


Fig. 2. Proposed RGT2TSKFNN structure.

process for each external input variable; $\upsilon_k(\tau)$ with internal recurrent variable with weight $\theta_k^{\prime}$ that can temporarily hold dynamic data and effectively manage temporal input noise. In this paper, a GT2FS is utilized with uncertain mean; $[M_1, M_2]$ and fixed width; $\circ$ to determine the lower membership function; $\underline{\psi}_{\prime k}^{(2)}$ and the upper membership function; $\overline{\psi}_{\prime k}^{(2)}$. The input to this layer i.e., membership layer can be expressed as:

$$\mathcal{I}_{\prime k}(\tau) = \upsilon_k(\tau) + \beta_k^{\prime}(\tau)\left(\frac{\overline{\psi}_{\prime k}^{(2)}(\tau-1) + \underline{\psi}_{\prime k}^{(2)}(\tau-1)}{2}\right) \tag{5}$$

where $\beta_k^{\prime}$ denotes the weight of the self-feedback loop. Furthermore, $\overline{\psi}_{\prime k}^{(2)}(\tau-1)$, and $\underline{\psi}_{\prime k}^{(2)}(\tau-1)$ are the previous output signals from layer 2 where it can be expressed as follows:

$$\underline{\psi}_{\prime k}^{(2)} = \begin{cases} exp\left\{\dfrac{-1}{2}\left(\dfrac{\mathcal{I}_{\prime k}(\tau) - M_{k2}^{\prime}}{\circ_k^{\prime}}\right)^2\right\}, \mathcal{I}_{\prime k}(\tau) \le \dfrac{M_{k1}^{\prime} + M_{k2}^{\prime}}{2} \\ exp\left\{\dfrac{-1}{2}\left(\dfrac{\mathcal{I}_{\prime k}(\tau) - M_{k1}^{\prime}}{\circ_k^{\prime}}\right)^2\right\}, \mathcal{I}_{\prime k}(\tau) > \dfrac{M_{k1}^{\prime} + M_{k2}^{\prime}}{2} \end{cases} \tag{6}$$

$$\overline{\psi}_{\prime k}^{(2)} = \begin{cases} exp\left\{\dfrac{-1}{2}\left(\dfrac{\mathcal{I}_{\prime k}(\tau) - M_{k1}^{\prime}}{\circ_k^{\prime}}\right)^2\right\}, & \mathcal{I}_{\prime k}(\tau) < M_{k1}^{\prime} \\ 1, M_{k1}^{\prime} \le \mathcal{I}_{\prime k}(\tau) \le M_{k2}^{\prime} \\ exp\left\{\dfrac{-1}{2}\left(\dfrac{\mathcal{I}_{\prime k}(\tau) - M_{k2}^{\prime}}{\circ_k^{\prime}}\right)^2\right\}, \mathcal{I}_{\prime k}(\tau) > M_{k2}^{\prime} \end{cases} \tag{7}$$

Consequently, the MF grade for each α -plane is determined by:

$$\mathcal{I}_{\prime k}^{\alpha_\vartheta} = \underline{\psi}_{\prime k}^{(2)} + 0.5 w_{\prime k}(\tau)\left[\overline{\psi}_{\prime k}^{(2)} - \underline{\psi}_{\prime k}^{(2)}\right]\alpha_\vartheta \tag{8}$$

$$\mathcal{J}_{\ell k}^{\alpha_\vartheta} = \overline{\psi}_{\ell k}^{(2)} - 0.5 w_{\ell k}(\tau)\left[\overline{\psi}_{\ell k}^{(2)} - \underline{\psi}_{\ell k}^{(2)}\right]\alpha_\vartheta \tag{9}$$

Layer 3 (Firing layer): In this layer, the number of nodes matches the number of recurrent rules, corresponding to the number of fuzzy sets in each input. Each recurrent rule node features an internal feedback loop that addresses uncertainties. The output of each node generates a temporal firing strength, which depends not only on the current spatial firing strength $\mathcal{F}_{\ell,\alpha_\vartheta}(\tau)$ but also on the previous temporal firing strength $\mathcal{F}_{\ell,\alpha_\vartheta}(\tau-1)$. The firing interval for each α-plane is assessed using an algebraic product operation as follows:

$$\mathcal{F}_{\ell,\alpha_\vartheta}(\tau) = \left[\underline{o}_{\ell,\alpha_\vartheta}(\tau), \overline{o}_{\ell,\alpha_\vartheta}(\tau)\right], \ell = 1, .., P \tag{10}$$

$$\underline{o}_{\ell,\alpha_\vartheta}(\tau) = \prod_{k=1}^{\chi} \mathcal{J}_{\ell k}^{\alpha_\vartheta}, \overline{o}_{\ell,\alpha_\vartheta}(\tau) = \prod_{k=1}^{\chi} \mathcal{J}_{\ell k}^{\alpha_\vartheta} \tag{11}$$

The output of layer three i.e., $\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}$ and $\overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)$ are calculated as:

$$\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau) = \Re^{\ell}(\tau)\underline{o}_{\ell,\alpha_\vartheta}(\tau) + \left(1 - \Re^{\ell}(\tau)\right)\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau-1) \tag{12}$$

$$\overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau) = \Re^{\ell}(\tau)\overline{o}_{\ell,\alpha_\vartheta}(\tau) + \left(1 - \Re^{\ell}(\tau)\right)\overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau-1) \tag{13}$$

where $\Re^{\ell}$ denotes an internal feedback weight, $(0 \le \Re^{\ell} \le 1)$. The value of $\Re^{\ell}$ represents the ratio of achievement between the current and previous inputs that influences the network output.

Layer 4 (Consequent layer): In this layer, the output of each node is represented as follows

$$\left[\underline{\psi}_{\ell}^{(4)}(\tau), \overline{\psi}_{\ell}^{(4)}(\tau)\right] = \left[a_0^{\ell} - b_0^{\ell}, a_0^{\ell} + b_0^{\ell}\right] + \sum_{k=1}^{\chi}\left[a_k^{\ell} - b_k^{\ell}, a_k^{\ell} + b_k^{\ell}\right]\mathcal{J}_{\ell k}(\tau) \tag{14}$$

Layer 5 (Type-reduction layer): The type-reduction process is performed using the μ_l and μ_r factors, which help to reduce the computational complexity of GT2FLS (El-Nagar et al., 2023). The result of the type-reduction process can be expressed as

$$\underline{\psi}_{\alpha_\vartheta}^{(5)}(\tau) = \frac{\left\{\mu_l \sum_{\ell=1}^{P} \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\,\underline{\psi}_{\ell}^{(4)}(\tau)\right\} + \left\{(1-\mu_l)\sum_{\ell=1}^{P}\overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\,\underline{\psi}_{\ell}^{(4)}(\tau)\right\}}{\sum_{\ell=1}^{P}\left(\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau) + \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\right)} \tag{15}$$

Furthermore,

$$\overline{\psi}_{\alpha_\vartheta}^{(5)}(\tau) = \frac{\left\{\mu_r \sum_{\ell=1}^{P} \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\;\overline{\psi}_{\ell}^{(4)}(\tau)\right\} + \left\{(1-\mu_r)\sum_{\ell=1}^{P}\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\;\overline{\psi}_{\ell}^{(4)}(\tau)\right\}}{\sum_{\ell=1}^{P}\left(\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau) + \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\right)} \tag{16}$$

where μ_l and μ_r denote design parameters.

Layer 6 (Output layer): Finally, the defuzzified output, $s(\tau)$ is determined in the sixth layer as follows

$$\psi_{\alpha_\vartheta}^{(6)}(\tau) = \underline{\psi}_{\alpha_\vartheta}^{(5)}(\tau) + \overline{\psi}_{\alpha_\vartheta}^{(5)}(\tau) \tag{17}$$

$$s(\tau) = \frac{\sum_{\vartheta=1}^{V} \alpha_\vartheta\, \psi_{\alpha_\vartheta}^{(6)}(\tau)}{\sum_{\vartheta=1}^{V}\alpha_\vartheta} \tag{18}$$

where V represents the total number of α-planes.

Remark 1. As highlighted in previous studies (Zhai and Mendel, 2011; Kumbasar and Hagras, 2014; Mohammadzadeh and Kaynak, 2019), a general T2-FLS (GT2-FLS) can be interpreted as an aggregation of multiple interval T2-FLS (IT2-FLSs), with each IT2-FLS corresponding to a specific value of α. Consequently, increasing the number of α-levels results in a greater number of embedded IT2-FLSs within the GT2-FLS, which enhances the system's performance and its ability to manage uncertainties. However, this improvement comes at the cost of increased computational complexity, as more α-levels directly translate to longer computation times. In this study, the selection of the number of α-levels was made by carefully balancing the trade-off between control performance and computational efficiency.

4. Online learning for RGT2TSKFNN

This section outlines the online structure and parameter learning for the proposed RGT2TSKFNN. Online structure learning is conducted to identify the number of rules in the network, while the parameter learning is carried out to update the network's parameters.

4.1. Online structure learning

In this section, the rules are derived from the measured data at each iteration. The proposed network employs fuzzy clustering to create fuzzy rules based on rule firing strength (Khater et al., 2019). The initial incoming data, $\upsilon_k(\tau)$, is utilized to generate the first type-2 fuzzy rule. For this first type-2 fuzzy rule, the initial parameters of the GT2FS are set as follows:

$$\left[M_{k1}^{1}, M_{k2}^{1}\right] = \left[\upsilon_k(\tau) - \varrho,\ \upsilon_k(\tau) + \varrho\ \right], \epsilon = \epsilon_{fixed}, k = 1, \ldots, \chi \tag{19}$$

where ϱ is a constant that indicates the interval range of the initial GT2FS. The value ϵ_{fixed} is a predefined value (in this study, we set it to 0.3). The average firing strength $\psi_{\ell,\alpha_\vartheta}(\tau)$ is calculated for each rule as follows:

$$\psi_{\ell,\alpha_\vartheta}(\tau) = \left(\underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau) + \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}(\tau)\right) \Big/ 2 \tag{20}$$

Next, for the following incoming data, $\upsilon_k(\tau)$, we define:

$$\rho = \underset{1 \le \ell \le P(\tau)}{argmax}\,\psi_{\ell,\alpha_\vartheta}(\tau) \tag{21}$$

A new rule is generated i.e., $P(\tau+1) = P(\tau) + 1$ if $\psi_{\rho,\alpha_\vartheta}(\tau) \le \psi_{th}$ (ψ_{th} denotes a predefined threshold) is met, where $P(\tau)$ represents the number of existing rules. The GT2FS parameters for each new rule are defined as follows:

$$\left[M_{k1}^{P(\tau)+1}, M_{k2}^{P(\tau)+1}\right] = \left[\upsilon_k(\tau) - \varrho,\ \upsilon_k(\tau) + \varrho\ \right], k = 1, \ldots, \chi \tag{22}$$

$$\epsilon^{P(\tau)+1} = \varepsilon\left|\upsilon_k(\tau) - \left(\frac{M_{k1}^{\rho} + M_{k2}^{\rho}}{2}\right)\right| \tag{23}$$

If the overlapping parameter, $\varepsilon = 0.5$, is present, it signifies that the new Gaussian GT2FS width is half of the Euclidean distance from the best matching center. The initial consequent parameters for a new rule are set as $a_0^{\ell} = 0.05$ and $b_0^{\ell} = 0.002$. The following algorithm illustrates the pseudo code for the online structure procedure implemented in the proposed RGT2TSKFNN discussed in this study.

Algorithm 1: The pseudo code of the structure learning for the suggested RGT2TSKFNN.

Initialization: $\tau = 1$.
Generate the first type-2 fuzzy rule; $P(\tau) = 1$.
  Set the initial values for the antecedent part of the RGT2TSKFNN given as in Eq. (19).
  Set the initial values for the consequent part as follows:
  For $k = 1$ *to* χ
  $a_0^1 = a_k^1 = 0.05$, and $b_k^1 = b_0^1 = 0.002$
  End For
While $\tau < N$ **do**
  If $\psi_{\rho,\alpha_\vartheta}(\tau) < \psi_{th}$

(*continued on next page*)

(*continued*)

Algorithm 1: The pseudo code of the structure learning for the suggested RGT2TSKFNN.

Generate a new rule; $P(\tau+1) = P(\tau)+1$.
Set the initial values for the uncertain mean and width as given in Eqs. (19) and (23).
Set the initial values for the consequent part as:
For $k = 1$ *to* χ
$a_0^{P(\tau)+1} = a_k^{P(\tau)+1} = 0.05$, and $b_k^{P(\tau)+1} = b_0^{P(\tau)+1} = 0.002$
End For
End If
End While

4.2. Online parameter learning

In this section, the Lyapunov function is utilized to update all the generated network parameters, and it is defined as follows:

$$\mathscr{H}(\tau) = \frac{1}{2}\left((e_h(\tau))^2 + (q(\tau))^2\right) \tag{24}$$

$$e_h(\tau) = s_d(\tau) - s(\tau) \tag{25}$$

where $\mathscr{H}(\tau)$ denotes a positive definite function, $e_h(\tau)$ denotes the error signal, $s(\tau)$ indicates the output of the RGT2TSKFNN, and $s_d(\tau)$ signifies the desired output. The parameters of the RGT2TSKFNN are denoted using $q(\tau)$.

Theorem. *The stability of the proposed RGT2TSKFNN is ensured through the Lyapunov function* $\mathscr{H}(\tau) = \frac{1}{2}\left((e_h(\tau))^2 + (q(\tau))^2\right) > 0$, *provided that the parameter update rule is established as follows*:

$$\Delta q(\tau) = -\frac{q(\tau) + e_h(\tau)\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)}{1 + \left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)^2} \tag{26}$$

Proof. *see Appendix*.

Following that, the updating parameters for the proposed RGT2TSKFNN are articulated as:

$$q(\tau+1) = q(\tau) - \eta\Delta q(\tau) = q(\tau) + \eta_q \frac{q(\tau) + e_h(\tau)\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)}{1 + \left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)^2} \tag{27}$$

where η_q *denotes the learning rate constant.*

4.2.1. Consequent parameters updating

By applying the aforementioned theorem, $\frac{\partial e_h(\tau)}{\partial q(\tau)}$ for the consequent parameters of the RGT2TSKFNN are derived as follows:

$$\frac{\partial e_h}{\partial a_0^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\frac{(1-\mu_l+\mu_r)\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + (1-\mu_r+\mu_l)\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}\right]\mathscr{I}_{\prime k}(\tau)\right\} \tag{28}$$

$$\frac{\partial e_h}{\partial b_0^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\frac{(1-\mu_l-\mu_r)\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} - (1-\mu_r-\mu_l)\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}\right]\mathscr{I}_{\prime k}(\tau)\right\} \tag{29}$$

$$\frac{\partial e_h}{\partial a_k^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\frac{(1-\mu_l+\mu_r)\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + (1-\mu_r+\mu_l)\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}\right]\mathscr{I}_{\prime k}(\tau)\right\} \tag{30}$$

$$\frac{\partial e_h}{\partial b_k^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\frac{(1-\mu_l-\mu_r)\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} - (1-\mu_r-\mu_l)\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}\right]\mathscr{I}_{\prime k}(\tau)\right\} \tag{31}$$

4.2.2. Antecedent parameters updating

By applying the aforementioned theorem, $\frac{\partial e_h(\tau)}{\partial q(\tau)}$ for the antecedent parameters of the RGIT2TSKFNN are derived as follows:

$$\frac{\partial e_h}{\partial M_{k1}^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k1}^{\prime}} + \left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k1}^{\prime}}\right]\right\} \tag{32}$$

$$\frac{\partial e_h}{\partial M_{k2}^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k2}^{\prime}} + \left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k2}^{\prime}}\right]\right\} \tag{33}$$

$$\frac{\partial e_h}{\partial \sigma_k^{\prime}} = \sum_{\vartheta=1}^{V}\left\{\frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V}\alpha_\vartheta}\left[\left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial\sigma_k^{\prime}} + \left(\frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} + \frac{\partial\psi_{\alpha_\vartheta}^{(6)}}{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}\right)\frac{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial\sigma_k^{\prime}}\right]\right\} \tag{34}$$

$$\frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} = \frac{(1-\mu_l)\underline{\psi}_{\prime}^{(4)} - \underline{\psi}_{\alpha_\vartheta}^{(5)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}, \frac{\partial\underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} = \frac{\mu_l\underline{\psi}_{\prime}^{(4)} - \underline{\psi}_{\alpha_\vartheta}^{(5)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)} \tag{35}$$

$$\frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} = \frac{\mu_r\overline{\psi}_{\prime}^{(4)} - \overline{\psi}_{\alpha_\vartheta}^{(5)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)}, \frac{\partial\overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}} = \frac{(1-\mu_r)\overline{\psi}_{\prime}^{(4)} - \overline{\psi}_{\alpha_\vartheta}^{(5)}}{\sum_{\prime=1}^{P}\left(\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)} + \overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}\right)} \tag{36}$$

$$\begin{aligned}\frac{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k1}^{\prime}} &= \frac{\partial\overline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial\overline{o}_{\prime,\alpha_\vartheta}}\frac{\partial\overline{o}_{\prime,\alpha_\vartheta}}{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}\left(\frac{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}{\partial\overline{\psi}_{\prime k}^{(2)}}\frac{\partial\overline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}} + \frac{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}{\partial\underline{\psi}_{\prime k}^{(2)}}\frac{\partial\underline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}}\right)\\ &= \left(\Re^{\prime}\prod_{i=1,i\neq k}^{\chi}\mathscr{J}_{\prime i}^{\alpha_\vartheta}\right)\left(\left[1-\frac{1}{2}w_{\prime k}\alpha_\vartheta\right]\frac{\partial\overline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}} + \left[\frac{1}{2}w_{\prime k}\alpha_\vartheta\right]\frac{\partial\underline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}}\right)\end{aligned} \tag{37}$$

$$\begin{aligned}\frac{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial M_{k1}^{\prime}} &= \frac{\partial\underline{\psi}_{\prime,\alpha_\vartheta}^{(3)}}{\partial\underline{o}_{\prime,\alpha_\vartheta}}\frac{\partial\underline{o}_{\prime,\alpha_\vartheta}}{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}\left(\frac{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}{\partial\underline{\psi}_{\prime k}^{(2)}}\frac{\partial\underline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}} - \frac{\partial\mathscr{J}_{\prime k}^{\alpha_\vartheta}}{\partial\overline{\psi}_{\prime k}^{(2)}}\frac{\partial\overline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}}\right)\\ &= \left(\Re^{\prime}\prod_{i=1,i\neq k}^{\chi}\mathscr{J}_{\prime i}^{\alpha_\vartheta}\right)\left(\left[1-\frac{1}{2}w_{\prime k}\alpha_\vartheta\right]\frac{\partial\underline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}} + \left[\frac{1}{2}w_{\prime k}\alpha_\vartheta\right]\frac{\partial\overline{\psi}_{\prime k}^{(2)}}{\partial M_{k1}^{\prime}}\right)\end{aligned} \tag{38}$$

$$\frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial M^{\ell}_{k2}} = \frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \overline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \overline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} + \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} \right) = \left(\Re^{\ell} \prod_{i=1, i\neq k}^{\chi} \mathscr{F}^{\alpha_\vartheta}_{\ell i} \right) \left(\left[1 - \frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} + \left[\frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} \right) \tag{39}$$

$$\frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial M^{\ell}_{k2}} = \frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \underline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \underline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} - \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} \right) = \left(\Re^{\ell} \prod_{i=1, i\neq k}^{\chi} \mathscr{F}^{\alpha_\vartheta}_{\ell i} \right) \left(\left[1 - \frac{1}{2} w \alpha_\vartheta \right] \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} + \left[\frac{1}{2} w \alpha_\vartheta \right] \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} \right) \tag{40}$$

$$\frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \sigma^{\ell}_{k}} = \frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \overline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \overline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} + \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} \right) = \left(\Re^{\ell} \prod_{i=1, i\neq k}^{\chi} \mathscr{F}^{\alpha_\vartheta}_{\ell i} \right) \left(\left[1 - \frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} + \left[\frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} \right) \tag{41}$$

$$\frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \sigma^{\ell}_{k}} = \frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \underline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \underline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} - \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} \right) = \left(\Re^{\ell} \prod_{i=1, i\neq k}^{\chi} \mathscr{F}^{\alpha_\vartheta}_{\ell i} \right) \left(\left[1 - \frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} + \left[\frac{1}{2} w_{\ell k} \alpha_\vartheta \right] \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} \right) \tag{42}$$

$$\frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k1}} = \begin{cases} \overline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1} \right)}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) < M^{\ell}_{k1} \\ 0, otherwise \end{cases} \tag{43}$$

$$\frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k1}} = \begin{cases} \underline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1} \right)}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) > \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \\ 0, otherwise \end{cases} \tag{44}$$

$$\frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k1}} = \begin{cases} \overline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2} \right)}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) > M^{\ell}_{k2} \\ 0, otherwise \end{cases} \tag{45}$$

$$\frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial M^{\ell}_{k2}} = \begin{cases} \underline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2} \right)}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) \leq \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \\ 0, otherwise \end{cases} \tag{46}$$

$$\frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} = \begin{cases} \overline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1} \right)^2}{\left(\sigma^{\ell}_{k} \right)^3} \right), \mathscr{I}_{\ell k}(\tau) < M^{\ell}_{k1} \\ 0, otherwise \\ \overline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2} \right)^2}{\left(\sigma^{\ell}_{k} \right)^3} \right), \mathscr{I}_{\ell k}(\tau) > M^{\ell}_{k2} \end{cases} \tag{47}$$

$$\frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \sigma^{\ell}_{k}} = \begin{cases} \underline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2} \right)^2}{\left(\sigma^{\ell}_{k} \right)^3} \right), \mathscr{I}_{\ell k}(\tau) \leq \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \\ \underline{\psi}^{(2)}_{\ell k} \left(\frac{\left(\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1} \right)^2}{\left(\sigma^{\ell}_{k} \right)^3} \right), \mathscr{I}_{\ell k}(\tau) > \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \end{cases} \tag{48}$$

4.2.3. Recurrent parameters updating

By applying the aforementioned theorem, $\frac{\partial e_h(\tau)}{\partial q(\tau)}$ for the recurrent parameters of the RGIT2TSKFNN are derived as follows:

$$\frac{\partial e_h}{\partial \Re^{\ell}} = \sum_{\vartheta=1}^{V} \left\{ \frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V} \alpha_\vartheta} \left[\left(\frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} + \frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} \right) \overline{o}_{\ell,\alpha_\vartheta} + \left(\frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} + \frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} \right) \underline{o}_{\ell,\alpha_\vartheta} \right] \right\} \tag{49}$$

$$\frac{\partial e_h}{\partial \beta^{\ell}_{k}} = \sum_{\vartheta=1}^{V} \left\{ \frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V} \alpha_\vartheta} \left[\left(\frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} + \frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} \right) \frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \beta^{\ell}_{k}} + \left(\frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \overline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} + \frac{\partial \psi^{(6)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}} \frac{\partial \underline{\psi}^{(5)}_{\alpha_\vartheta}}{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}} \right) \frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\beta^{\ell}_{k}} \right] \right\} \tag{50}$$

$$\frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \beta^{\ell}_{k}} = \frac{\partial \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \overline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \overline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} + \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} \right) \tag{51}$$

$$\frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \beta^{\ell}_{k}} = \frac{\partial \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta}}{\partial \underline{o}_{\ell,\alpha_\vartheta}} \frac{\partial \underline{o}_{\ell,\alpha_\vartheta}}{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}} \left(\frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \underline{\psi}^{(2)}_{\ell k}} \frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} + \frac{\partial \mathscr{F}^{\alpha_\vartheta}_{\ell k}}{\partial \overline{\psi}^{(2)}_{\ell k}} \frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} \right) \tag{52}$$

$$\frac{\partial \overline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} = \begin{cases} \overline{\psi}^{(2)}_{\ell k} \left(\frac{\overline{\psi}^{(2)}_{\ell k}(\tau-1) + \underline{\psi}^{(2)}_{\ell k}(\tau-1)}{2} \right) \left(\left(\frac{\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1}}{\left(\sigma^{\ell}_{k} \right)^2} \right) \right), \mathscr{I}_{\ell k}(\tau) < M^{\ell}_{k1} \\ 0, otherwise \\ \overline{\psi}^{(2)}_{\ell k} \left(\frac{\overline{\psi}^{(2)}_{\ell k}(\tau-1) + \underline{\psi}^{(2)}_{\ell k}(\tau-1)}{2} \right) \left(\left(\frac{\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2}}{\left(\sigma^{\ell}_{k} \right)^2} \right) \right), \mathscr{I}_{\ell k}(\tau) > M^{\ell}_{k2} \end{cases} \tag{53}$$

$$\frac{\partial \underline{\psi}^{(2)}_{\ell k}}{\partial \beta^{\ell}_{k}} = \begin{cases} \underline{\psi}^{(2)}_{\ell k} \left(\frac{\overline{\psi}^{(2)}_{\ell k}(\tau-1) + \underline{\psi}^{(2)}_{\ell k}(\tau-1)}{2} \right) \left(\frac{\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k2}}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) \leq \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \\ \underline{\psi}^{(2)}_{\ell k} \left(\frac{\overline{\psi}^{(2)}_{\ell k}(\tau-1) + \underline{\psi}^{(2)}_{\ell k}(\tau-1)}{2} \right) \left(\frac{\mathscr{I}_{\ell k}(\tau) - M^{\ell}_{k1}}{\left(\sigma^{\ell}_{k} \right)^2} \right), \mathscr{I}_{\ell k}(\tau) > \frac{M^{\ell}_{k1} + M^{\ell}_{k2}}{2} \end{cases} \tag{54}$$

4.2.4. Type-reduction parameters updating

By applying the aforementioned theorem, $\frac{\partial e_h(\tau)}{\partial q(\tau)}$ for the type-reduction parameters $[\mu_l, \mu_r]$ of the RGIT2TSKFNN are derived as follows:

$$\frac{\partial e_h}{\partial \mu_l} = \sum_{\vartheta=1}^{V} \left\{ \frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V} \alpha_\vartheta} \left[\frac{\sum_{\ell=1}^{P} \underline{\psi}^{(4)}_{\ell} \left(\underline{\psi}^{(3)}_{\ell,\alpha_\vartheta} - \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta} \right)}{\sum_{\ell=1}^{P} \left(\underline{\psi}^{(3)}_{\ell,\alpha_\vartheta} + \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta} \right)} \right] \right\} \tag{55}$$

$$\frac{\partial e_h}{\partial \mu_r} = \sum_{\vartheta=1}^{V} \left\{ \frac{\alpha_\vartheta}{\sum_{\vartheta=1}^{V} \alpha_\vartheta} \left[\frac{\sum_{\ell=1}^{P} \overline{\psi}^{(4)}_{\ell} \left(\overline{\psi}^{(3)}_{\ell,\alpha_\vartheta} - \underline{\psi}^{(3)}_{\ell,\alpha_\vartheta} \right)}{\sum_{\ell=1}^{P} \left(\underline{\psi}^{(3)}_{\ell,\alpha_\vartheta} + \overline{\psi}^{(3)}_{\ell,\alpha_\vartheta} \right)} \right] \right\} \tag{56}$$

4.2.5. The parameter of secondary MF

Using the previously mentioned theorem, $\frac{\partial e_h(\tau)}{\partial q(\tau)}$ for the parameter w, which control the shape of the secondary membership function can be expressed as follows:

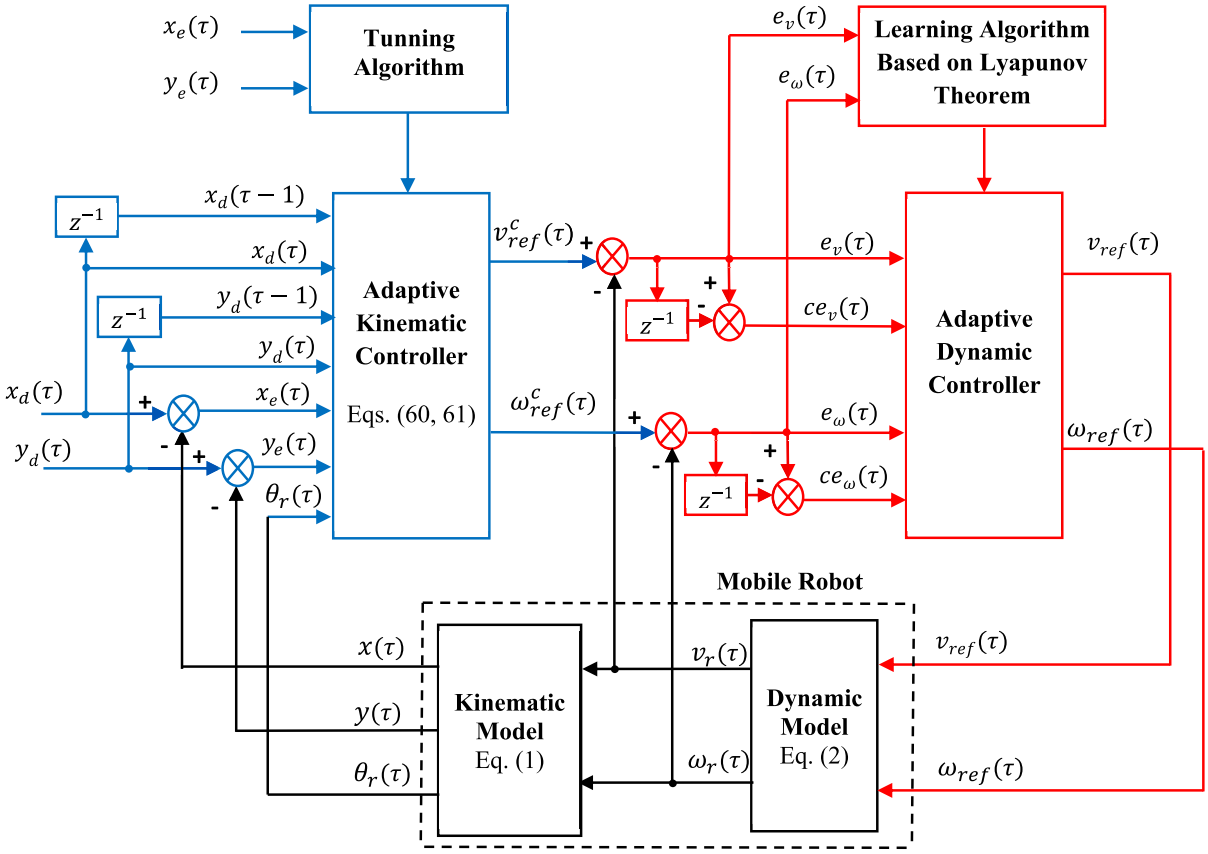


Fig. 3. Block diagram of the proposed trajectory tracking controller.

$$\frac{\partial e_h}{\partial w_{\ell k}} = \frac{\partial e_h}{\partial s}\sum_{\vartheta=1}^{V}\left\{\frac{\partial s}{\partial \psi_{\alpha_\vartheta}^{(6)}}\left[\left(\frac{\partial \psi_{\alpha_\vartheta}^{(6)}}{\partial \underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial \underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial \overline{\psi}_{\alpha_\vartheta}^{(3)}} + \frac{\partial \psi_{\alpha_\vartheta}^{(6)}}{\partial \overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial \overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial \overline{\psi}_{\alpha_\vartheta}^{(3)}}\right)\frac{\partial \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial w_{\ell k}} + \left(\frac{\partial \psi_{\alpha_\vartheta}^{(6)}}{\partial \underline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial \underline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}} + \frac{\partial \psi_{\alpha_\vartheta}^{(6)}}{\partial \overline{\psi}_{\alpha_\vartheta}^{(5)}}\frac{\partial \overline{\psi}_{\alpha_\vartheta}^{(5)}}{\partial \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}\right)\frac{\partial \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial w_{\ell k}}\right]\right\} \tag{57}$$

$$\frac{\partial \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial w_{\ell k}} = \frac{\partial \overline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial \mathscr{I}_{\ell k}^{\alpha_\vartheta}}\frac{\partial \mathscr{I}_{\ell k}^{\alpha_\vartheta}}{\partial w_{\ell k}} = 0.5\Re^{\vee}\left[\overline{\psi}_{\ell k}^{(2)} - \underline{\psi}_{\ell k}^{(2)}\right]\alpha_\vartheta \prod_{i=1, i\neq k}^{\chi}\mathscr{I}_{\ell i}^{\alpha_\vartheta} \tag{58}$$

$$\frac{\partial \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial w_{\ell k}} = \frac{\partial \underline{\psi}_{\ell,\alpha_\vartheta}^{(3)}}{\partial \mathscr{I}_{\ell k}^{\alpha_\vartheta}}\frac{\partial \mathscr{I}_{\ell k}^{\alpha_\vartheta}}{\partial w_{\ell k}} = 0.5\Re^{\vee}\left[\overline{\psi}_{\ell k}^{(2)} - \underline{\psi}_{\ell k}^{(2)}\right]\alpha_\vartheta \prod_{i=1, i\neq k}^{\chi}\mathscr{I}_{\ell i}^{\alpha_\vartheta} \tag{59}$$

5. Design of trajectory tracking controller

Fig. 3 illustrates the block diagram of the proposed controller for the mobile robot, which is composed of three main components. The first component represents the mobile robot itself, with two inputs: the desired linear velocity $v_{ref}(\tau)$ and the desired angular velocity $\omega_{ref}(\tau)$. This block produces three outputs: the robot's x and y positions, as well as its orientation θ_r. The mathematical equations describing this component are presented in Section 2. The second part is the adaptive kinematic controller, whose objective is to generate the reference signals for the linear and angular velocities, denoted as $v_{ref}^c(\tau)$ and $\omega_{ref}^c(\tau)$, respectively. The mathematical formulation of this controller is detailed in Section 5.1. The third part is the adaptive dynamic controller, which produces the actual control signals, linear and angular velocities, to be applied to the mobile robot. The mathematical equations describing this part are provided in Section 5.2.

5.1. Adaptive kinematic controller

The adaptive kinematic controller is designed based on a kinematic model of the mobile robot. The proposed adaptive kinematic controller is given as (Martins et al., 2008):

$$v_{ref}^c(\tau) = \left[\left(\frac{x_d(\tau) - x_d(\tau-1)}{T}\right) + l_x(\tau)\ tanh\left(\frac{g_x(\tau)}{l_x(\tau)}\ x_e(\tau)\right)\right]\cos(\theta_r(\tau)) + \left[\left(\frac{y_d(\tau) - y_d(\tau-1)}{T}\right) + l_y(\tau)\ tanh\left(\frac{g_y(\tau)}{l_y(\tau)}\ y_e(\tau)\right)\right]\sin(\theta_r(\tau)) \tag{60}$$

$$\omega_{ref}^c(\tau) = \left[\left(\frac{x_d(\tau) - x_d(\tau-1)}{T}\right) + l_x(\tau)\ tanh\left(\frac{g_x(\tau)}{l_x(\tau)}x_e(\tau)\right)\right]\left(\frac{-1}{a_r}\sin(\theta_r(\tau))\right) + \left[\left(\frac{y_d(\tau) - y_d(\tau-1)}{T}\right) + l_y(\tau)\ tanh\left(\frac{g_y(\tau)}{l_y(\tau)}y_e(\tau)\right)\right]\left(\frac{1}{a_r}\cos(\theta_r(\tau))\right) \tag{61}$$

where $x_e(\tau) = x_d(\tau) - x(\tau)$ and $y_e(\tau) = y_d(\tau) - y(\tau)$ are current position errors in x and y positions, respectively. $x_d(\tau)$ and $y_d(\tau)$ are the desired positions in x and y positions, respectively. $x(\tau)$ and $y(\tau)$ are the current positions in x and y positions, respectively. $g_x(\tau) > 0$ and $g_y(\tau) > 0$ are the controller gains, while $l_x(\tau) \in \mathbb{R}$ and $l_y(\tau) \in \mathbb{R}$ are the saturation constants. In this paper, the controller gains and constants are updated based on the following proposed algorithm:

Step 1: Determine the relative error as:

$$e_r(\tau) = \frac{RMSE(\tau) - RMSE(\tau-1)}{RMSE(\tau)} \tag{62}$$

where $RMSE(\tau)$ is the root mean square error, which is used in this paper

as a cost function. It is defined as:

$$RMSE(\tau)=\sqrt{\frac{1}{N}\sum_{\tau=1}^{N}\left(x_e^2(\tau)+y_e^2(\tau)\right)} \tag{63}$$

Step 2: The proposed adjustment of the controller gains and saturation constants can be made as follows:

Case 1. for $e_r(\tau)<0$

$$g_x(\tau+1)=g_x(\tau)\left[1+\mu_{gx}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{gx}\in[0,1] \tag{64}$$

$$g_y(\tau+1)=g_y(\tau)\left[1+\mu_{gy}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{gy}\in[0,1] \tag{65}$$

$$l_x(\tau+1)=l_x(\tau)\left[1+\mu_{lx}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{lx}\in[0,1] \tag{66}$$

$$l_y(\tau+1)=l_y(\tau)\left[1+\mu_{ly}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{ly}\in[0,1] \tag{67}$$

Case 2. for $e_r(\tau)\geq 0$

$$g_x(\tau+1)=g_x(\tau)\left[1-\mu_{gx}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{gx}\in[0,1] \tag{68}$$

$$g_y(\tau+1)=g_y(\tau)\left[1-\mu_{gy}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{gy}\in[0,1] \tag{69}$$

$$l_x(\tau+1)=l_x(\tau)\left[1-\mu_{lx}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{lx}\in[0,1] \tag{70}$$

$$l_y(\tau+1)=l_y(\tau)\left[1-\mu_{ly}\,exp\left(\frac{-e_r(\tau)}{2}\right)\right],\mu_{ly}\in[0,1] \tag{71}$$

5.2. Adaptive dynamic controller

The adaptive dynamic controller is designed using the proposed RGT2TSKFNN, presented in section 2. There are two networks: the first one is for the linear velocity, with inputs for this network being the error in linear velocity, $e_v(\tau)$, and the change of the error in linear velocity, $ce_v(\tau)=e_v(\tau)-e_v(\tau-1)$, and the output is $v_{ref}(\tau)$. The firing threshold value for the linear velocity network is set to $\psi_{th}=0.25$, where it influences and decides the number of fuzzy rules to be generated. To trade-off between the structure complexity and the ability to improve the controller against uncertainties, three α-planes are used for each GT2FS. Choosing the right learning rate for the proposed network is critical for both convergence speed and performance. Therefore, to trade-off between the convergence speed and performance, the learning rates for this network are set to $\eta_a=0.4$, $\eta_b=0.2$, $\eta_{M_1}=0.2$, $\eta_{M_2}=0.01$, $\eta_c=0.05$, $\eta_{\Re}=0.1$, $\eta_\theta=0.01$, $\eta_{\mu_l}=0.15$; $\eta_{\mu_r}=0.1$ and $\eta_w=0.2$. The error signal $e_h(\tau)$, which is used in Eq. (24) for this network is $e_v(\tau)$.

The second network is for the angular velocity, with inputs for this network are the error in angular velocity, $e_\omega(\tau)$, and the change of the error in angular velocity, $ce_\omega(\tau)=e_\omega(\tau)-e_\omega(\tau-1)$, and the output is $\omega_{ref}(\tau)$. The firing threshold value for the angular velocity network is set to $\psi_{th}=0.001$ and the number of α-planes is selected as 3. The learning rates for this network are set to $\eta_a=0.04$, $\eta_b=0.03$, $\eta_{M_1}=0.15$, $\eta_{M_2}=0.05$, $\eta_c=0.01$, $\eta_{\Re}=0.1$, $\eta_\theta=0.01$, $\eta_{\mu_l}=0.1$; $\eta_{\mu_r}=0.05$ and $\eta_w=0.2$. The error signal $e_h(\tau)$, used in Eq. (24) for this network is $e_\omega(\tau)$.

6. Simulation results

The trajectory tracking controller based on RTSKIT2FNN proposed in (Bencherif and Chouireb, 2019) was applied to the same mobile robot (Pioneer DX3). The results were compared with other networks published previously such as ANFIS (Aissa and Fatima, 2015), RFNN (Mon et al., 2008), IT2FNN-SM (Kayacan and Maslim, 2016), IT2FNN (Abiyev and Kaynak, 2010), self-organization T2FNN using gradient descent for parameters learning and the particle swarm optimization (Lin and Le, 2017), SEIT2FNN (Lin et al., 2018) and SIT2FNN (Lin et al., 2013). The obtained results in (Bencherif and Chouireb, 2019) based on RTSKIT2FNN are better than other networks and RTSKIT2FNN can reduce the obtained fuzzy rules. Therefore, the simulation results for the developed adaptive tracking controller based on the proposed RGT2TSKFNN are compared with the tracking controller using RTSKIT2FNN (Bencherif and Chouireb, 2019) and self-organizing interval type-2 fuzzy neural network (SOIT2FNN) based on fractional order sliding mode control (Zhao et al., 2024).

To measure the performance of each network, the performance indices such as mean average error (MAE), integral of absolute error (IAE) and integral of square error (ISE) are considered in this study. These performance indices are defined as (El-Nagar, 2016; Sakr et al., 2018; Shaheen et al., 2020):

$$MAE=\frac{1}{N}\sum_{\tau=1}^{N}|x_e(\tau)|+|y_e(\tau)| \tag{72}$$

$$IAE=\int_0^t\left(|x_e(\tau)|+|y_e(\tau)|\right)dt \tag{73}$$

$$ISE=\int_0^t\left(x_e^2(\tau)+y_e^2(\tau)\right)\,dt \tag{74}$$

On the other hand, the percentage improvement (PI) in tracking is calculated as the relative difference between the values for each scheme (Mohareri et al., 2012). As an example, the MAE improvement index is given as:

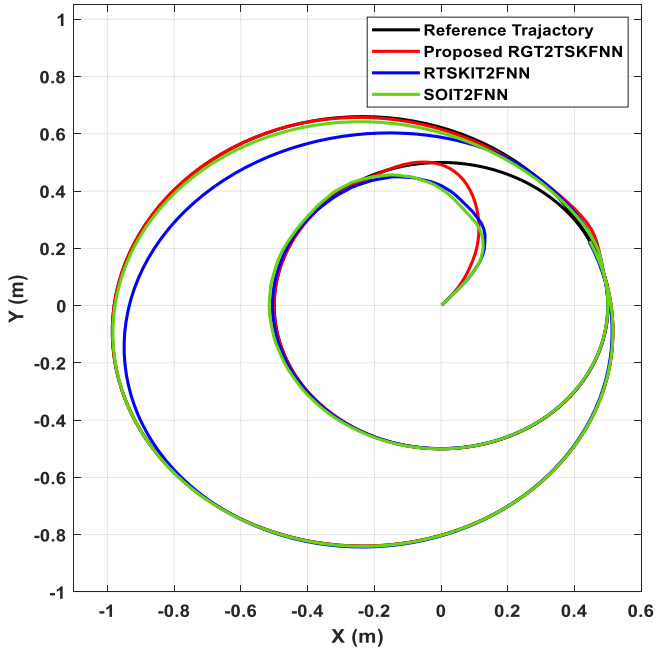


Fig. 4. Tracking response of the mobile robot with a reference circular trajectory (Task 1 – Case 1).

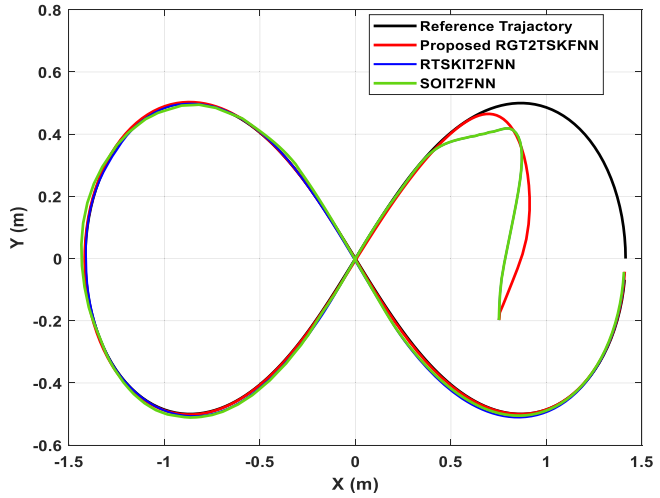


Fig. 5. Tracking response of the mobile robot with a reference Lemniscate trajectory (Task 1 – Case 2).

$$PI = \frac{MAE_{RTSKIT2FNN} - MAE_{RGT2TSKFNN}}{MAE_{RTSKIT2FNN}} \times 100 \tag{75}$$

6.1. Task 1: Tracking reference trajectory

In this task, the proposed controller is tested by applying various reference trajectories to demonstrate its robustness in ensuring that the mobile robot accurately follows a reference. Two cases are considered in this task as follows:

Case 1. Circle Trajectory

In this scenario, the mobile robot is required to follow a reference circular trajectory, initially starting at $x = 0.0\,m$ and $y = 0.0\,m$. Fig. 4 shows the reference trajectory and the actual trajectory of the mobile robot on the X-Y plane, using the proposed RGT2TSKFNN controller as well as two benchmark controllers: RTSKIT2FNN and SOIT2FNN. The black line represents the reference trajectory, the red line represents the trajectory obtained using the proposed RGT2TSKFNN controller, the blue line corresponds to the trajectory using RTSKIT2FNN and the green line shows the trajectory using SOIT2FNN. As depicted in Fig. 4, the initial reference trajectory is a circle with a radius of 0.5 m. After 13 s, the reference trajectory transitions to a larger circle with a radius of 0.75 m, starting at $x = 0.5\,m$ and $y = 0.0\,m$.

Case 2. Lemniscate Trajectory

In this case, the mobile robot is tasked with following a reference Lemniscate trajectory, starting from the initial position $x = 0.75\,m$ and $y = -0.2\,m$. Fig. 5 shows both the reference and the measured

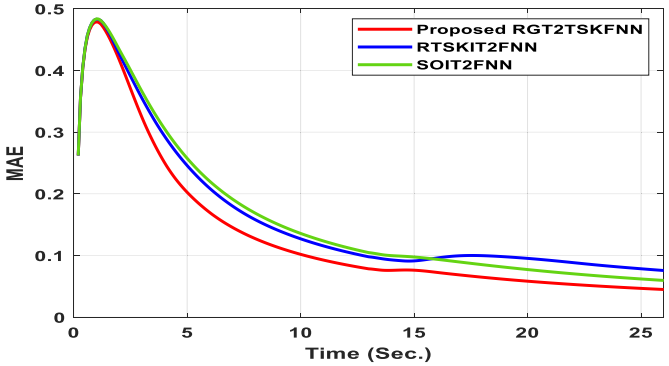


Fig. 6. MAE for both controllers (Task 1 – Case 1).

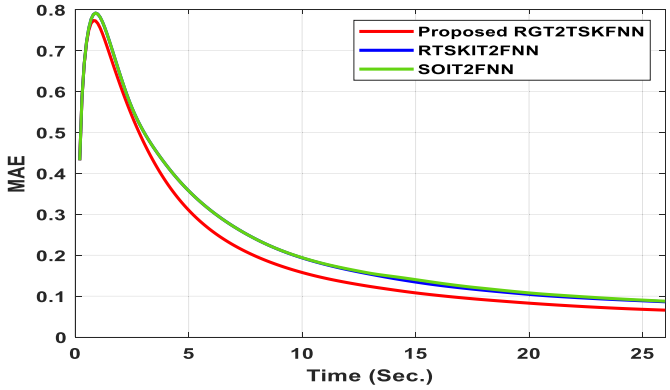


Fig. 7. MAE for both controllers (Task 1 – Case 2).

trajectories of the mobile robot on the X–Y plane using the proposed RGT2TSKFNN controller and two comparative controllers: RTSKIT2FNN and SOIT2FNN. The reference trajectory starts at $x = 1.4\,m$ and $y = 0.0\,m$. The figure highlights the ability of each controller to track the complex Lemniscate path under the given initial conditions.

As illustrated in Figs. 4 and 5, the trajectory of the mobile robot using the developed adaptive tracking controller based on the proposed RGT2TSKFNN closely follows the reference trajectory, outperforming both the RTSKIT2FNN and SOIT2FNN controllers. This observation is further supported by the results shown in Figs. 6 and 7, which present the MAE for each controller. The proposed RGT2TSKFNN-based controller consistently achieves a lower MAE compared to the RTSKIT2FNN and SOIT2FNN controllers, indicating improved tracking accuracy. These results confirm that the developed adaptive tracking controller based on the proposed RGT2TSKFNN significantly enhances the trajectory tracking performance of the mobile robot.

6.2. Task 2: Applying variable disturbances

In this task, the proposed controller is evaluated under varying disturbances, denoted as ρ_x, ρ_y, ρ_v and ρ_ω to assess its robustness and ability to handle uncertainties arising from such disturbances. Two test cases

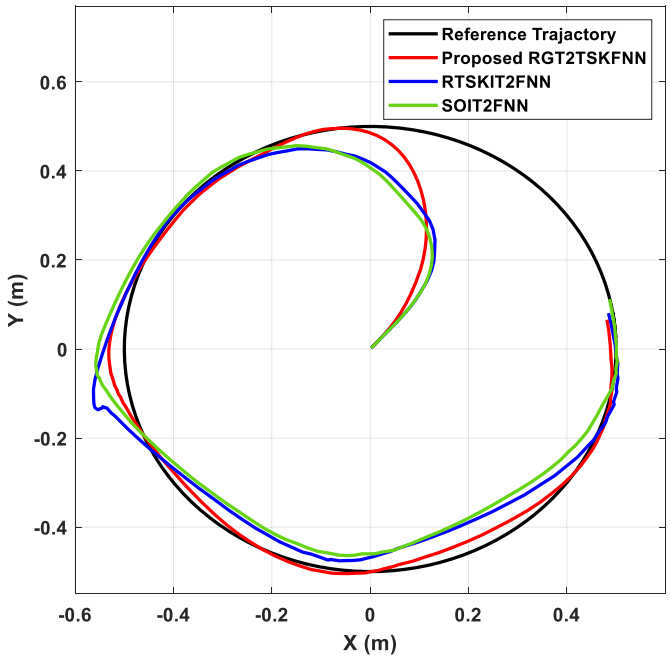


Fig. 8. Tracking response of the mobile robot with a reference circle trajectory with disturbance (Task 2 – Case 1).

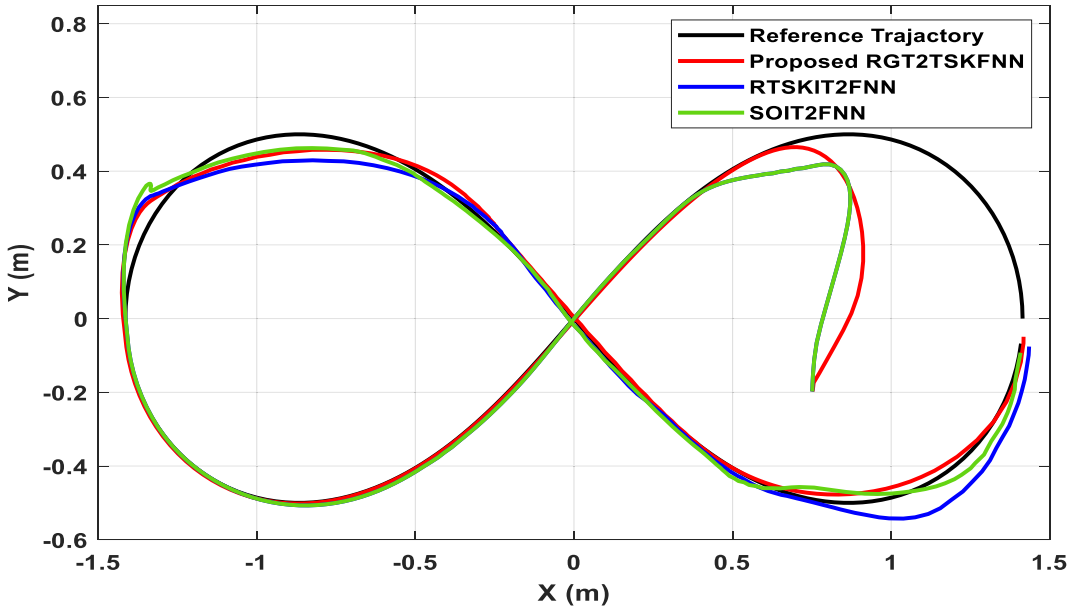


Fig. 9. Tracking response of the mobile robot with a reference Lemniscate trajectory with disturbance (Task 2 – Case 2).

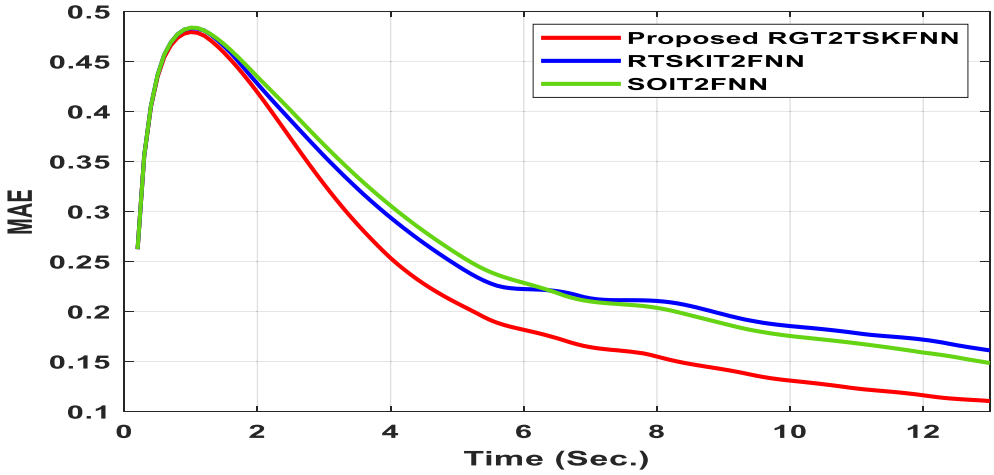


Fig. 10. MAE for both controllers (Task 2 – Case 1).

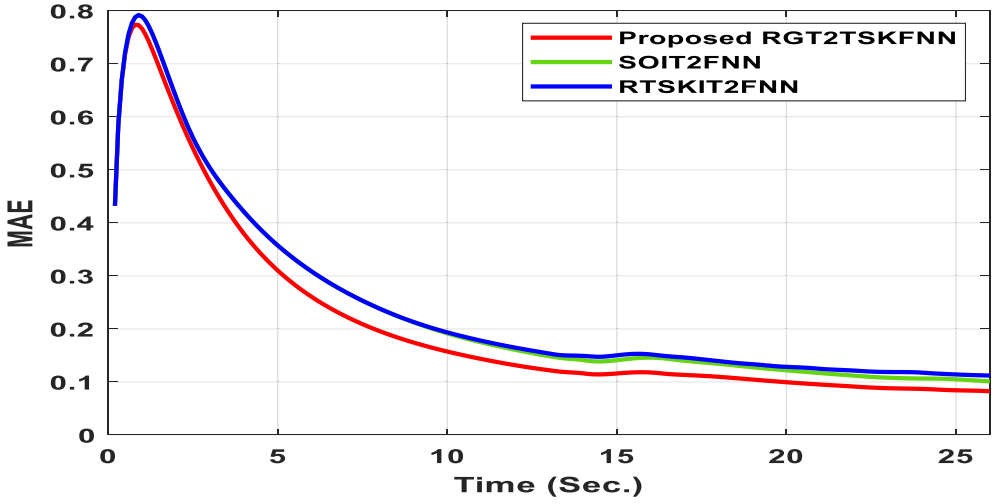


Fig. 11. MAE for both controllers (Task 2 – Case 2).

are considered in this task, as described below:

Case 1: in this case, the reference trajectory is a circle having radius equal 0.5 m, which starts at $x = 0.5\ m$ and $y = 0.0\ m$. The mobile robot starts at $x = 0.0\ m$ and $y = 0.0\ m$. The variable disturbance values; $\rho_x = -0.01\sin(0.25\,\pi\,t)\sin(\theta_r)$, $\rho_y = 0.01\sin(0.25\,\pi\,t)\cos(\theta_r)$, $\rho_v = 0.15\sin(0.75\,\pi\,t) + 0.15\cos(0.75\,\pi\,t)$ and $\rho_\omega = 0.15\cos(0.75\,\pi\,t) - 0.15\sin(0.75\,\pi\,t)$ are applied on mobile robot after 5 s.

Case 2: in this second case, the reference trajectory is Lemniscate, that starts at $x = 1.4\ m$ and $y = 0.0\ m$. The mobile robot starts at $x = 0.75\ m$ and $y = -0.2\ m$. the disturbance values $\rho_x = -0.01\sin(0.25\,\pi\,t)\sin(\theta_r)$, $\rho_\omega = 0.01\sin(0.25\,\pi\,t)\cos(\theta_r)$, $\rho_v = 0.1\sin(0.75\,\pi\,t) + 0.1\cos(0.75\,\pi\,t)$ and $\rho_2 = 0.1\cos(0.75\,\pi\,t) - 0.1\sin(0.75\,\pi\,t)$ are applied on mobile robot after 13 s.

Figs. 8 and 9 present the reference trajectory alongside the actual trajectory of the mobile robot in the X–Y plane for both Case 1 and Case 2, respectively. These figures clearly demonstrate that the proposed controller outperforms the RTSKIT2FNN and SOIT2FNN controllers in the presence of variable disturbances. Figs. 10 and 11 further support this observation by showing the MAE for each controller. The proposed RGT2TSKFNN-based controller consistently achieves a lower MAE compared to the other networks. These results confirm that the developed adaptive tracking controller based on the proposed RGT2TSKFNN effectively mitigates the impact of variable external disturbances and enhances overall trajectory tracking performance.

6.3. Task 3: Effect of parameter uncertainties

In this task, the proposed controller is evaluated under parameter uncertainties introduced into the dynamic model of the mobile robot, in order to assess its robustness to model variations. Two test cases are considered, as outlined below:

Case 1. in this case, 60 % parameter uncertainties are applied with circle reference trajectory after 5 s.

Case 2. in this case, 53 % parameter uncertainties are applied with Lemniscate reference trajectory after 13 s.

Figs. 12 and 13 illustrate the reference trajectory and the actual trajectory of the mobile robot in the X–Y plane for both test cases

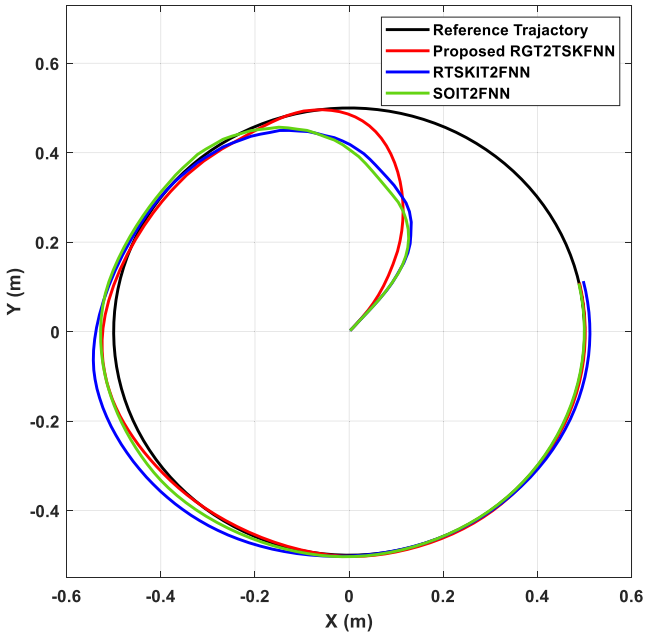


Fig. 12. Tracking response of the mobile robot with a reference circle trajectory with parameter uncertainties (Task 3 – Case 1).

involving parameter uncertainties. The results clearly show that the proposed controller maintains superior tracking performance, with its trajectory remaining closer to the reference path compared to the RTSKIT2FNN and SOIT2FNN controllers. This improved performance is further confirmed by the MAE values presented in Figs. 14 and 15, where the proposed RGT2TSKFNN-based controller consistently achieves lower MAE than the other controllers under parameter uncertainty conditions.

Tables 1–3 present a performance comparison among the proposed RGT2TSKFNN controller and the benchmark controllers, RTSKIT2FNN and SOIT2FNN, across the three tasks discussed above. The proposed network consistently achieves lower values of performance indices, the IAE, ISE, and MAE, compared to the other controllers. Additionally, the number of fuzzy rules required for both linear and angular velocity control in the proposed RGT2TSKFNN is lower than that of the RTSKIT2FNN and SOIT2FNN controllers, indicating improved structural efficiency. As summarized in Tables 4–6, the developed adaptive tracking controller based on the proposed RGT2TSKFNN effectively enhances trajectory tracking performance while reducing the impact of disturbances and model uncertainties, all while maintaining acceptable PI.

Remark 2. Monitoring the evolution of the tracking error over iterations or time provides valuable insights into the convergence behavior of the learning algorithm and can be used to detect potential learning saturation (Ahmadi et al., 2024; Kontoni and Ahmadi, 2024). Accordingly, Fig. 16 presents the tracking errors in the x and y positions over time for the proposed RGT2TSKFNN in simulation Task 1, Case 1. This example demonstrates that the proposed network achieves rapid learning saturation.

Remark 3. To illustrate the online learning behavior and structural adaptation capability of the proposed controller, Fig. 17 depicts the evolution of the number of fuzzy rules for both the linear and angular velocity controllers over time, using Simulation Task 1 – Case 1 (Circle Trajectory) as an example. The figure provides a clear visualization of the structure learning process, where it is evident that the number of rules changes dynamically in response to modifications in the reference trajectory specifically, when the trajectory transitions to a circle with a larger radius.

7. Experimental results

To evaluate the robustness and the performance of the developed adaptive tracking controller based on the proposed RGT2TSKFNN in real-world conditions, the Pioneer 3-DX mobile robot shown in Fig. 18 is used in this paper. The Pioneer 3-DX mobile robot is an all-purpose base, used for research and many applications involving autonomous navigation, path planning, mapping, teleoperation, and other behaviors. It has two driven 20 cm wheels with a passive caster wheel and encoders for measuring wheel rotation and tracking movement. The robot can reach speeds of up to 1.2 m/s and is equipped with an internal-based PC for processing and running algorithms.

Fig. 19 shows the performance of the robot for both controllers while tracking the circular trajectory having a radius of 0.7 m. After 20 s, the reference trajectory changes to a circle with a larger radius equal 0.9 m, starting at $x = 0.7\ m$ and $y = 0.0\ m$. The trajectory of the mobile robot using the developed adaptive tracking controller based on the proposed RGT2TSKFNN is very close to the reference trajectory compared with the RTSKIT2FNN and SOIT2FNN controllers.

Fig. 20 illustrates the reference and actual trajectories for the circular path under the influence of uncertainties in position (x–y), linear velocity, and angular velocity measurements. The results demonstrate that the proposed adaptive tracking controller, based on the RGT2TSKFNN framework, effectively mitigates the impact of these uncertainties compared to other controllers.

Table 7 presents a comparative analysis of the proposed

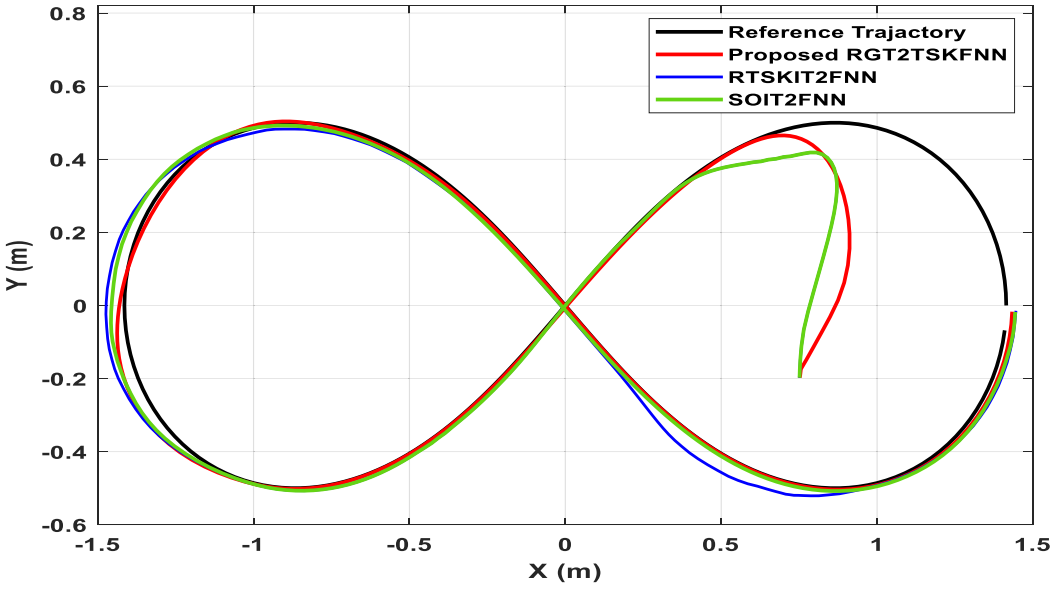


Fig. 13. Tracking response of the mobile robot with a reference Lemniscate trajectory with parameter uncertainties (Task 3 – Case 2).

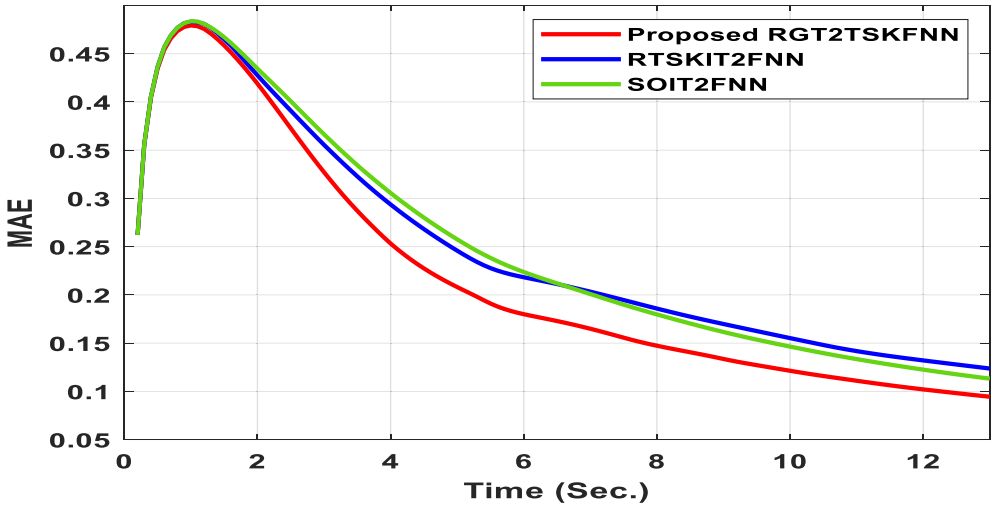


Fig. 14. MAE for both controllers (Task 3 – Case 1).

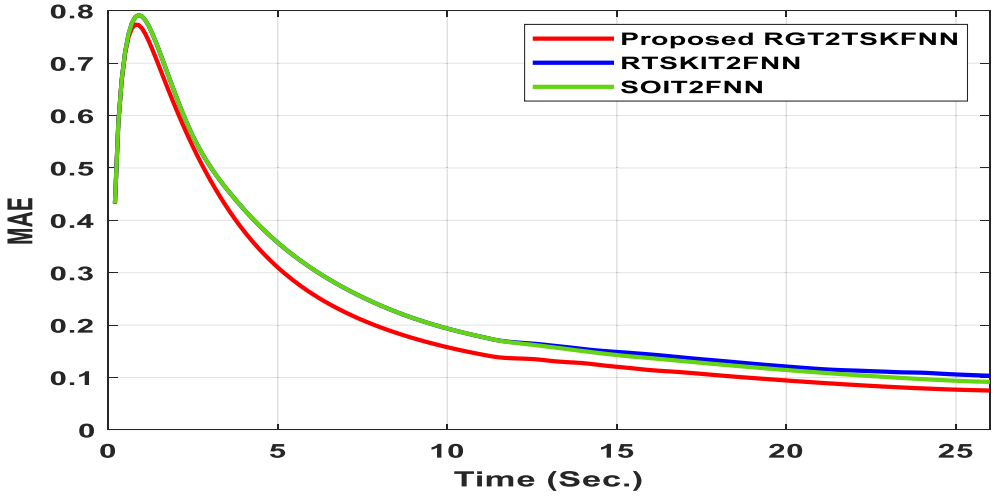


Fig. 15. MAE for both controllers (Task 3 – Case 2).

Table 1
Comparison results for Task 1.

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN	Proposed RGT2TSKFNN
Case 1: Circle Trajectory	IAE	19.67	15.49	11.67
	ISE	3.508	3.218	2.926
	MAE	0.0757	0.0596	0.0449
	Rule number (v)	9	7	6
	Rule number (ω)	5	5	4
Case 2: Lemniscate Trajectory	IAE	22.42	22.81	17.11
	ISE	6.238	6.250	5.450
	MAE	0.0863	0.0877	0.0658
	Rule number (v)	2	2	2
	Rule number (ω)	7	6	4

Table 2
Comparison results for Task 2.

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN	Proposed RGT2TSKFNN
Case 1:	IAE	20.93	19.28	14.34
	ISE	3.578	3.542	3.033
	MAE	0.1610	0.1483	0.1103
	Rule number (v)	12	9	8
	Rule number (ω)	3	3	2
Case 2:	IAE	26.30	25.10	21.44
	ISE	7.059	6.735	6.017
	MAE	0.1012	0.0966	0.0825
	Rule number (v)	4	4	4
	Rule number (ω)	11	9	4

Table 3
Comparison results for Task 3.

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN	Proposed RGT2TSKFNN
Case 1:	IAE	16.07	15.457	12.28
	ISE	3.207	3.152	2.941
	MAE	0.1237	0.1191	0.0945
	Rule number (v)	8	7	5
	Rule number (ω)	3	3	2
Case 2:	IAE	26.87	26.10	19.53
	ISE	6.491	6.301	5.570
	MAE	0.1034	0.1004	0.0751
	Rule number (v)	2	2	2
	Rule number (ω)	8	7	4

Table 4
Improvement ratios (%) between the proposed network and other models (Task 1).

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN
Case 1: Circle Trajectory	IAE	40.1	24.7
	ISE	16.6	9.1
	MAE	40.1	24.7
	Rule number (v)	33.3	14.3
	Rule number (ω)	20	20
Case 2: Lemniscate Trajectory	IAE	23.8	25.0
	ISE	12.6	12.8
	MAE	23.8	25.0
	Rule number (v)	0	0
	Rule number (ω)	42.9	33.3

Table 5
Improvement ratios (%) between the proposed network and other models (Task 2).

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN
Case 1:	IAE	31.5	25.6
	ISE	15.2	14.4
	MAE	31.5	25.6
	Rule number (v)	33.3	11.1
	Rule number (ω)	33.3	33.3
Case 2:	IAE	18.5	14.6
	ISE	14.8	10.6
	MAE	18.5	14.6
	Rule number (v)	0	0
	Rule number (ω)	63.6	55.6

Table 6
Improvement ratios (%) between the proposed network and other models (Task 3).

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN
Case 1:	IAE	23.6	20.6
	ISE	8.3	6.7
	MAE	23.6	20.6
	Rule number (v)	37.5	28.6
	Rule number (ω)	33.3	33.3
Case 2:	IAE	27.4	25.2
	ISE	14.2	11.6
	MAE	27.4	25.2
	Rule number (v)	0	0
	Rule number (ω)	50	42.9

RGT2TSKFNN controller against the RTSKIT2FNN and SOIT2FNN controllers for the experimental tasks described above. The performance metrics: IAE, ISE, and MAE for the proposed RGT2TSKFNN are consistently lower than those of the RTSKIT2FNN controller, with values averaged over 15 trials. These results indicate that the adaptive tracking controller based on the RGT2TSKFNN framework enhances the trajectory tracking performance of the mobile robot and effectively mitigates the impact of measurement uncertainties, while maintaining acceptable PI, as detailed in Table 8.

8. Discussion and comparative analysis

Based on the preceding comparative analysis, the following key findings are presented:

1) *The proposed adaptive tracking controller, developed based on the RGT2TSKFNN framework, significantly enhances the trajectory tracking performance of the mobile robot.* Simulation results (Task 1) along with the data in Tables 1 and 4 demonstrate notable improvements in IAE, ISE, and MAE, achieving acceptable PI in comparison with other controllers. Similarly, the experimental results, as reported in Tables 7 and 8, confirm the controller's effectiveness in reducing IAE, ISE, and MAE under real-world conditions, further validating its robustness and reliability.
2) The proposed trajectory tracking controller demonstrates enhanced robustness in handling various sources of uncertainty, including variable disturbances, parameter variations, and measurement errors, through the integration of fuzzy neural networks based on GT2FSs.
   - In the presence of variable disturbances, the proposed controller achieves improved IAE, ISE, and MAE values, with acceptable performance indices, compared to other controllers, as evidenced in Tables 2 and 5

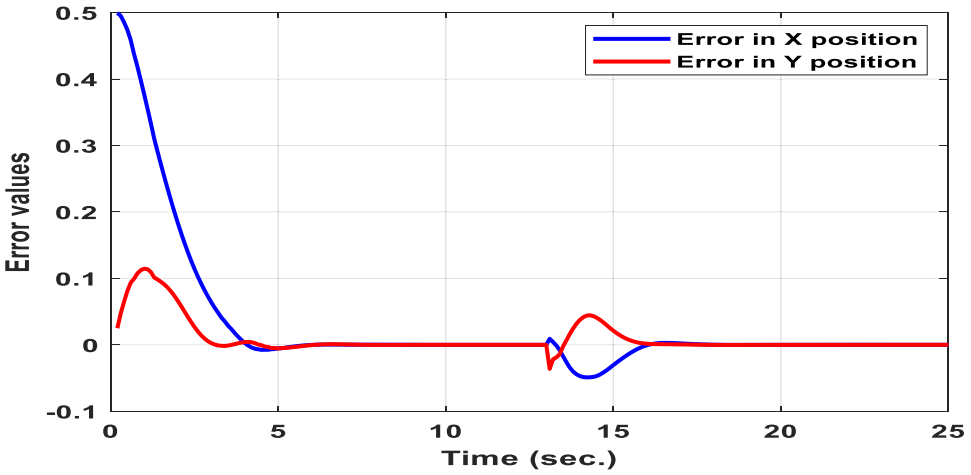


Fig. 16. Tracking error in x and y positions (Task 1 – Case 1).

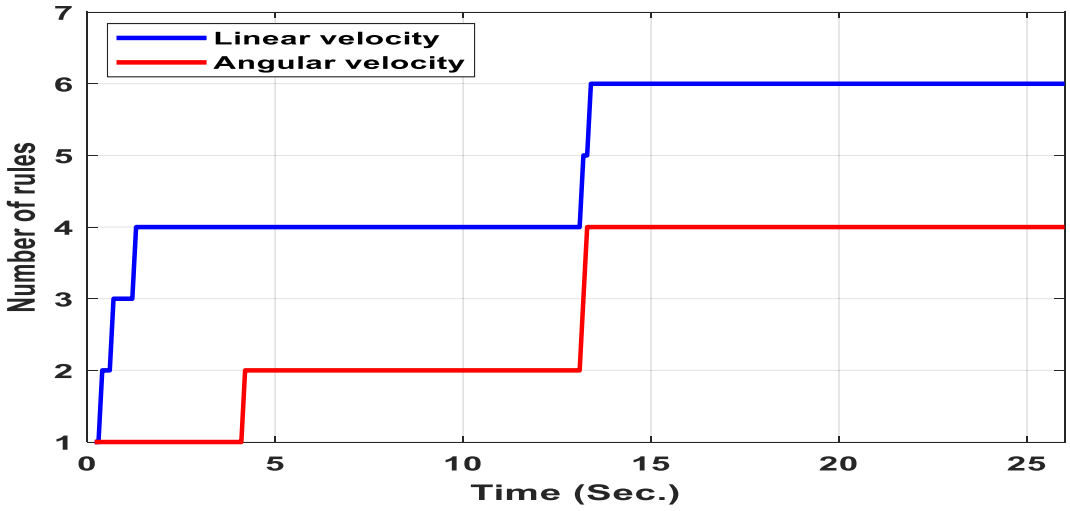


Fig. 17. Number of rules for linear and angular velocities for the proposed controller (Task 1 – Case 1).



Fig. 18. Pioneer 3DX mobile robot.

- When subjected to parameter variations, the controller maintains superior performance in terms of IAE, ISE, and MAE, as shown in Tables 3 and 6
- Under measurement errors, the controller continues to outperform its counterparts, as demonstrated by the improved error metrics in Tables 7 and 8

These results confirm the controller's capacity to manage different types of uncertainties effectively.

3) The proposed network architecture effectively reduces the number of required fuzzy rules, thereby minimizing structural complexity while maintaining high performance.

As shown in Tables 1–3, the proposed approach significantly reduces the size of the fuzzy rule base compared to conventional methods. This reduction contributes to enhanced computational efficiency and simplified implementation, without degrading the controller's accuracy or robustness.

4) The stability of the proposed adaptive trajectory tracking controller is ensured through the derivation of parameter update laws based on Lyapunov stability theory.

By formulating the adaptation mechanism according to Lyapunov criteria, the proposed controller guarantees convergence and

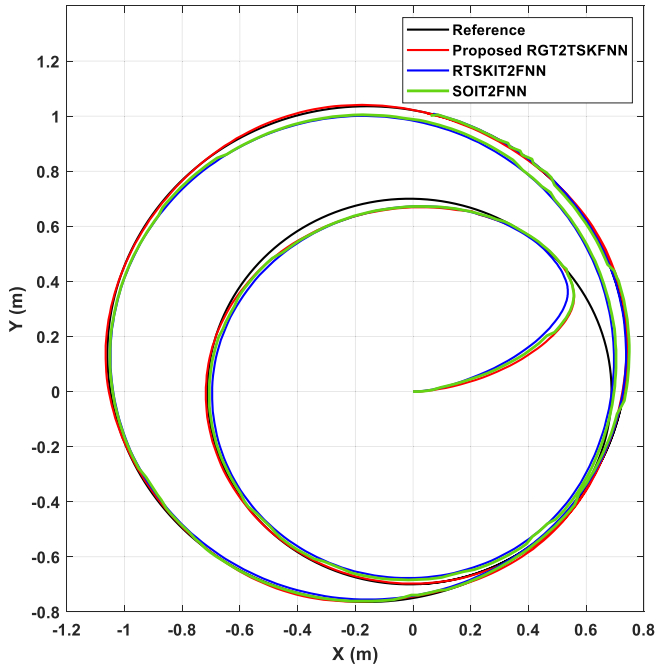


Fig. 19. Real trajectory of the mobile robot with a reference circular trajectory.

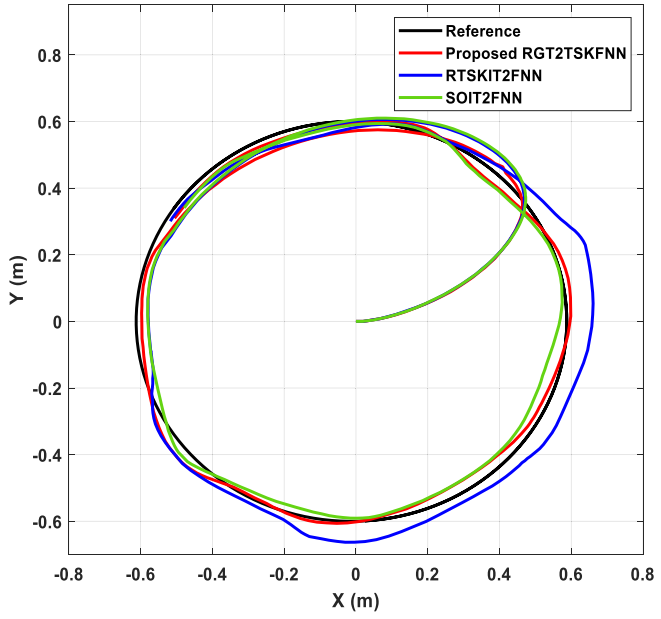


Fig. 20. Real trajectory of the mobile robot with a reference circular trajectory under applying uncertainties in measurements.

Table 7
Comparison results for experimental tasks.

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN	Proposed RGT2TSKFNN
Circular trajectory	IAE	52.21	51.51	45.63
	ISE	16.79	16.69	16.24
	MAE	0.1044	0.1031	0.0913
Circular trajectory with uncertainties	IAE	43.24	42.31	39.66
	ISE	1.328	1.258	1.087
	MAE	0.1730	0.1695	0.1587

Table 8
Improvement ratios (%) between the proposed network and other models for experimental tasks.

Tasks	Performance Index	RIT2TSKFNN	SOIT2FNN
Circular trajectory	IAE	12.6	11.4
	ISE	3.3	2.7
	MAE	12.6	11.4
Circular trajectory with uncertainties	IAE	8.3	6.4
	ISE	18.2	13.6
	MAE	8.3	6.4

system stability during trajectory tracking under varying uncertainties.

Table 9 presents a comparative analysis between the proposed controller and previously published controllers applied to mobile robots. Based on this comparison, along with the results obtained from both simulation and experimental tasks, the main advantages of the proposed adaptive trajectory tracking controller, developed using the RGT2TSKFNN framework, can be summarized as follows:

1) *Reduced rule base complexity:* The network architecture, which incorporates a RFNN with TSK type rules, significantly reduces the number of required fuzzy rules, thereby enhancing computational efficiency.
2) *Guaranteed stability:* The parameter update laws are derived based on Lyapunov stability theory, ensuring the controller's stable performance under varying conditions.
3) *Robustness to uncertainties and disturbances:* The GT2FS in the input membership functions enables the controller to effectively manage system uncertainties and environmental disturbances.

These findings highlight the proposed controller's superior tracking performance, structural efficiency, and robustness, making it well-suited for real-world mobile robotic applications.

9. Conclusions

This paper presents an adaptive tracking controller for a mobile robot, consisting of two principal components: an adaptive kinematic controller and an adaptive dynamic controller. The kinematic controller dynamically adjusts its gains online to ensure precise trajectory tracking, while the dynamic controller, developed using the proposed RGT2TSKFNN, effectively compensates for uncertainties caused by variable disturbances, parameter variations, and measurement errors. The RGT2TSKFNN is trained online through simultaneous structure and parameter learning, with stability rigorously guaranteed via the Lyapunov stability method, which provides the update laws for parameter adaptation.

The proposed controller is implemented and validated experimentally on a Pioneer 3DX mobile robot. Comparative tests against previously applied controllers; RIT2TSKFNN and SOIT2FNN, demonstrate the superiority of the RGT2TSKFNN-based controller. The robustness of the proposed approach is further evaluated under various uncertainties, including external disturbances, parameter variations, and measurement errors. Results indicate that the proposed controller reduces network complexity by minimizing the number of fuzzy rules generated. Moreover, it significantly improves trajectory tracking performance and effectively mitigates the impact of uncertainties, as reflected by reduced performance indices such as IAE, ISE, and MAE.

This work contributes to the field by providing a reliable control solution for mobile robots operating under challenging and uncertain conditions. However, a notable limitation of the current study is the increased computational complexity associated with the use of GT2FSs. There exists an inherent trade-off between performance and

Table 9
Comparison between all algorithms.

Algorithms	Structure learning	Parameter Learning	Type of fuzzy sets	Structure of rules	Simulation/ Experimental	Uncertainty Test	Stability Analysis
IT2FNN (Kim and Chwa, 2014)	Fixed	–	IT2FS	Mamdani	Both	NO	NO
ANFIS (Aissa and Fatima, 2015)	Fixed	Gradient Method	T1FS	TSK	Simulation	NO	NO
RTSKIT2FNN (Bencherif and Chouireb, 2019)	Learned	Gradient Method	IT2FS	TSK	Experimental	NO	Lyapunov
Adaptive PID Hanna et al. (2023)	–	Lyapunov	–	–	Experimental	YES	Lyapunov
IT2-FLC Al-Mahturi et al. (2023)	Learned	Sliding Surface	IT2FS	TSK	Simulation	YES	Lyapunov
SOIT2FNN Zhao et al. (2024)	Learned	Lyapunov	IT2FS	Mamdani	Both	YES	Lyapunov
Proposed controller	Learned	Lyapunov	GT2FS	TSK	Both	YES	Lyapunov

computational load: increasing the number of α-planes enhances controller performance but also raises computation time, and vice versa. In this study, this trade-off was addressed by selecting three α-plane values based on designer expertise. Future work will focus on developing an optimization algorithm to determine the optimal number of α-planes for GT2FSs, thereby balancing performance and computational efficiency more systematically. Additionally, the proposed approach will be extended to accommodate multi-robot systems, broadening its applicability and study the effect of varied surface conditions.

CRediT authorship contribution statement

Ahmad M. El-Nagar: Writing – original draft, Validation, Software, Methodology, Formal analysis. **Raouf Fareh:** Writing – review & editing, Project administration. **Sofiane Khadraoui:** Writing – review & editing, Visualization. **Maamar Bettayeb:** Visualization, Validation. **Tamer Rabie:** Writing – review & editing, Visualization. **Ibrahim Kamel:** Writing – review & editing, Visualization. **A. Aziz Khater:** Writing – review & editing, Software, Methodology, Formal analysis.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT to only check and improve the language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Let the variation in the Lyapunov function be designated as:

$$\Delta\mathscr{K}(\tau)=\mathscr{K}(\tau+1)-\mathscr{K}(\tau)=\frac{1}{2}\left\{\left[(e_h(\tau+1))^2+(q(\tau+1))^2\right]-\left[(e_h(\tau))^2+(q(\tau))^2\right]\right\} \tag{A1}$$

Eq. (A1) can be rewritten as:

$$\Delta\mathscr{K}(\tau)=\frac{1}{2}\{[(e_h(\tau+1)+e_h(\tau))\,(e_h(\tau+1)-e_h(\tau))\,]+[(q(\tau+1)+q(\tau))\,(q(\tau+1)-q(\tau))\,]\} \tag{A2}$$

Let $\Delta e_h(\tau)=e_h(\tau+1)-e_h(\tau)$ and $(\tau)=q(\tau+1)-q(\tau)$, Eq. (A2) can be rewritten as:

$$\Delta\mathscr{K}(\tau)=\frac{1}{2}\{\Delta e_h(\tau)(\Delta e_h(\tau)+2e_h(\tau))+\Delta q(\tau)(\Delta q(\tau)+2q(\tau))\} \tag{A3}$$

$$\Delta\mathscr{K}(\tau)=\frac{1}{2}\left\{(\Delta e_h(\tau))^2\left[1+\left(\frac{\Delta e_h(\tau)}{\Delta q(\tau)}\right)^2\right]+2\Delta q(\tau)\left[q(\tau)+e_h(\tau)\left(\frac{\Delta e_h(\tau)}{\Delta q(\tau)}\right)\right]\right\} \tag{A4}$$

The Lyapunov stability proof is valid locally for small variations of $\frac{\Delta e_h(\tau)}{\Delta q(\tau)}$, thus Eq. (A4) can be rewritten as:

$$\Delta\mathscr{K}(\tau)=\frac{1}{2}\left\{(\Delta e_h(\tau))^2\left[1+\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)^2\right]+2\Delta q(\tau)\left[q(\tau)+e_h(\tau)\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)\right]\right\} \tag{A5}$$

The second condition, represented by $\Delta\mathscr{K}(\tau)\leq 0$, is required to achieve stability. Consequently:

$$\Delta\mathscr{K}(\tau)=\frac{1}{2}\left\{(\Delta e_h(\tau))^2\left[1+\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)^2\right]+2\Delta q(\tau)\left[q(\tau)+e_h(\tau)\left(\frac{\partial e_h(\tau)}{\partial q(\tau)}\right)\right]\right\}\leq 0 \tag{A6}$$

The stability condition is satisfied after performing certain mathematical procedures on Eq. (A6).

Data availability

No data was used for the research described in the article.

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